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1 Executive Summary

This report presents the findings from comprehensive applied field and laboratory research studies conducted with scaffolders and carpenters from 15/09/2025 to 15/03/2026. The purpose is to evaluate the biomechanical demands during simulated scaffolding assembly and disassembly tasks, which involve repetitive hammering performed on elevated platforms. Twenty participants completed the study: 10 participants were assessed under controlled laboratory conditions and 10 participants were assessed in the field. In the laboratory, participants performed a typical scaffolding task, informed by task observation and subject matter experts, consisting of: (1) scaffold pole adjustment/de-adjustment [POLE], (2) hammering into the bottom knob of a scaffold structure [FLOOR], and (3) hammering onto a formwork wall knob [WALL]. In the field, participants performed striking tasks on two surface types (scaffold fitting/knob and wood). Across both lab and field scenarios, participants completed the tasks using four scaffold hammers: Estwing 16 oz (H16), Estwing 20 oz (H20), Estwing 22 oz (H22), and Stiletto 15 oz (HS). During all trials, muscle activity, shoulder and elbow joint range of motion, and applied force were recorded. Through a combination of task observations, posture assessments, and physiological measurements, this report provides a comprehensive, evidence-informed understanding of the biomechanical demands associated with the four hammer designs. From these data, practical recommendations for risk reduction while hammering and performance optimization are identified.

Key Findings



Field

- **Field study finding: Surface type mattered more than hammer type for shoulder and arm effort. Striking scaffold fittings/knobs generally required higher muscle effort and larger joint movement than striking wood.**
- **Both the 16 oz and 20 oz Estwing hammers reduced most muscle and movement demands when hammering on wood surfaces. With these hammers, there was a ~75% reduction in shoulder abduction/adduction range of motion when hammering wood (a more compliant surface) versus the knob surface.**
- **On the knob surface, the 22 oz Estwing (H22) showed the lowest average exertive force.** It ranked highest for “efficiency” (lowest mean EMG across the measured muscles) and also ranked highest for “effectiveness” (more joint motion per unit of muscle effort).
- **On wood, H16 and H22 performed similarly for efficiency, with H22 still ranking highest for effectiveness.**
- **The standard hammer (HS) produced the highest forearm extensor loading.** This was most notable on wood (mean 32.8% MVC) and was statistically higher than the other hammer conditions ($p = 0.026$).
- **Peak impact forces were higher on the knob surface for H20 and H22.** Compared to HS and H16, hammering with H20 and H22 produced significantly higher peak impact forces ($p < 0.0001$), while vibration transmission across hammer types were not meaningfully different. Generally, heavier hammers increased impact forces on hard fittings but did not clearly attenuate vibration.



Laboratory

- **Task condition influenced movement demands more than hammer type.** Across the simulated conditions, working height/target location drove the largest changes in shoulder and elbow range of motion.
- **Floor-level striking produced the greatest shoulder flexion demands.** Compared to wall and pole conditions, floor striking consistently required larger shoulder flexion ranges, indicating higher postural demand in that scenario.
- **Differences between hammers were generally small and inconsistent across tasks.** While some hammers were ranked higher for specific performance metrics (e.g., efficiency/effectiveness) within certain conditions, overall patterns did not show a single hammer outperforming all others across laboratory tasks.
- **No clear differences between hammers based on both muscle activity and range of motion.** Within-condition tests were not significant for the primary EMG and ROM measures, suggesting any observed differences should be interpreted as descriptive within the current sample.

Implications and Recommendations

Based on the field findings, both the strike surface and hammer type influenced the exertive forces required by a worker. The recommendations below focus on possible hazard controls that partners can pilot and refine in their own environments.

- **For knob-surface work, consider replacing the standard hammer (HS) with H22 (preferred) or H20.** Field results suggest H22 required lower overall muscle demands when hammering on the knob surface while still delivering effective strikes. *Why this matters:* HS also showed the highest forearm extensor activation, which may increase fatigue during repeated use.
- **When task conditions permit, prioritize wood-surface striking and pair it with lighter hammers (H16 or H20).** Field results showed substantially lower muscle demands required when hammering on wood surfaces compared with knob surfaces. *Why this matters:* planning the work so the strike target is more compliant may reduce overall arm and shoulder exertive forces.

Collectively, these strategies may enhance worker safety, health, and performance.

Based on the laboratory findings, posture and task setup (e.g., strike height and target location) are primary drivers of shoulder and elbow muscle loading. Any differences between hammer types are



generally smaller and task-dependent. Recommendations below emphasize controlling the work configuration first before considering the brand or specification of hammer.

- **Control strike height and reach first (engineering/work setup).** Floor-level striking consistently increased shoulder flexion demands; where feasible, raise the work/strike point to reduce sustained shoulder flexion and awkward trunk/arm postures.
- **Consider hammer brand and specification once tasks and workspaces are ergonomically designed.** Laboratory results suggest hammer-to-hammer differences were smaller than task effects and therefore both tasks and workspaces should be ergonomically designed first to achieve full benefits of hammer effects; selection should prioritize usability, control, and fit for the specific task configuration.
- **When impact is required (e.g., stiffer targets), consider a heavier hammer option.** Heavier hammers may improve strike delivery in harder contact conditions, potentially reducing the need for repeated swings.
- **Use administrative controls for high-repetition striking scenarios.** Apply task rotation and planned micro-breaks when repetitive striking cannot be eliminated, especially for floor-level or otherwise constrained postures.



General recommendations (Field + Laboratory)

- **Prioritize controlling the strike condition before relying on tool choice.** Where possible, change the strike target or setup first (e.g., avoid hard, low-compliance contact when an alternative is available and change working heights to minimize excessive reaching).
- **Match hammer type to surface/target demands.** Field results suggest surface type plays a significant role in muscle loading; consider heavier options for hard fitting/knob work and lighter options where the target is more compliant, when the task allows.
- **Reduce cumulative exposure in repetitive striking tasks.** Use task rotation, pacing, and scheduled breaks to manage fatigue, even when single-strike muscle demand appears lower.
- **Standardize a short “tool + task setup” check before high-volume striking.** Confirm handle condition/grip, clear strike line-of-sight, stable stance on elevated platforms, and optimal working height to minimize awkward shoulder positions.
- **Pilot changes and monitor outcomes.** Test preferred hammer(s) and setup changes with a small group, then track comfort/fatigue feedback and any quality/throughput impacts to support continuous improvement.



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5 List of Abbreviations

Abbreviation	Definition
ADELTA	Anterior Deltoid
MDELTA	Middle Deltoid
PDELTA	Posterior Deltoid
BICEP	Biceps Brachii
FOREARM	Forearm Extensor
EMG	Electromyography
MVC	Maximum Voluntary Contraction
ROM	Range of Motion
Sh_ABD_ADD	Shoulder Abduction/Adduction
Sh_ER_IR	Shoulder External/Internal Rotation
Sh_FL_EXT	Shoulder Flexion/Extension
EL_FL_EXT	Elbow Flexion/Extension
HS	Stiletto 15 oz Hammer (control condition)
H16	Estwing 16 oz Hammer
H20	Estwing 20 oz Hammer
H22	Estwing 22 oz Hammer
SD	Standard Deviation
ANOVA	Analysis of Variance
MSD	Musculoskeletal Disorder
PPE	Personal Protective Equipment
HAV	Hand-Arm Vibration



6 Background and Objective

6.1 Injury Context and Motivation

Musculoskeletal disorders (MSDs) of the shoulder, elbow, and wrist/forearm are among the most common and costly occupational injuries in construction. Workers who perform repetitive hammer striking tasks are at an elevated risk due to sustained awkward postures, repetitive forceful exertions, excessive gripping, and impact forces. In scaffolding work, these demands are often amplified because assembly and disassembly require extensive hammering on elevated platforms, where stability, posture, and tool interaction directly affect both safety and performance.

Hammer design is a modifiable risk factor that may alter the biomechanical demands of striking tasks. Head weight, handle geometry, material, and balance characteristics can influence the force required, grip mechanics, upper-extremity muscle activation patterns, and vibration transmitted to the hand and arm (Haque, 2018). Hammering also involves repetitive, multi-joint movements with variable shoulder, elbow, and wrist postures, contributing to cumulative strain and fatigue over time (Dupuis et al., 2021). Despite these known relationships, no studies have assessed how hammer design affects biomechanical demands specifically within scaffolding tasks, representing a critical gap in the literature.

6.2 Purpose

The purpose of this study was to evaluate the biomechanical and ergonomic demands associated with repetitive hammering during representative scaffolding assembly and disassembly tasks performed on elevated platforms. Using both laboratory and field assessments, the study compared four hammers used for scaffolding tasks (Estwing 16 oz [H16], Estwing 20 oz [H20], Estwing 22 oz [H22], and Stiletto 15 oz [HS]) across two strike surfaces in the field (scaffold fitting/knob and wood) and 3 scaffold assembling conditions in the laboratory ((1) scaffold pole adjustment/de-adjustment [POLE], (2) hammering into the bottom knob of a scaffold structure [FLOOR], and (3) hammering onto a formwork wall knob [WALL]), by measuring muscle activity, shoulder and elbow joint range of motion, applied/impact force. The specific aims of this report were to:

1. Quantify upper-extremity muscle activation (EMG) and joint range of motion (ROM) during active striking phases across hammer × surface conditions.
2. Measure impact force at the tool–surface interface and describe related force transmission characteristics.
3. Compare conditions to identify hammer designs and work conditions that reduce biomechanical demands while maintaining effective task performance.
4. Provide practical, evidence-informed recommendations for hammer selection and task design in scaffolding and related construction work.

This study also tested the following hypotheses: (1) shoulder and elbow ROM and movement strategies would differ across hammer designs; (2) muscle loading in the shoulder, upper arm, and forearm would vary by hammer type; and (3) impact force at the tool-surface and force transmission to the forearm would vary by hammer type and condition.



6.3 Description of Occupation and Study Context

Participants were experienced scaffolders and carpenters who regularly perform repetitive striking tasks as part of scaffolding assembly/disassembly and related construction work. Study tasks were selected to reflect common real-world scenarios involving elevated working positions and repeated hammer strikes. To balance realism and control, data were collected in both laboratory and field settings.

In the field, two strike surface conditions were examined:

- **Knob surface condition:** striking a scaffold fitting/knob (metal-to-metal contact) representative of scaffold connections during assembly/disassembly; this is a hard, low-compliance surface that typically requires precise, forceful strikes.
- **Wood surface condition:** striking wood representative of nailing/fastening operations (e.g., formwork or framing); this is a more compliant surface with greater energy absorption.

In the laboratory, the following conditions were examined:

- **Scaffold pole adjustment/de-adjustment** - involving rotational and percussive actions to loosen and secure scaffold connections;
- **Hammering into the bottom knob of a scaffold structure** - simulating forceful metal-to-metal impacts commonly required during scaffold assembly and disassembly;
- **Hammering onto a formwork wall knob** - representing fastening tasks against a semi-rigid support.

Across both settings, the same hammer set was used (Estwing 16 oz, 20 oz, 22 oz, and Stiletto 15 oz) to support consistent comparison across conditions.



7 Methods

This study used a combination of laboratory testing and applied field measurement to quantify the biomechanical demands of repetitive hammering during representative scaffolding assembly and disassembly tasks performed on elevated platforms (15/09/2025 to 15/03/2026). Twenty participants completed the study (10 laboratory; 10 field). In the laboratory, participants performed three simulated scaffolding conditions: (1) scaffold pole adjustment/de-adjustment (POLE), (2) hammering into the bottom knob of a scaffold structure (FLOOR), and (3) hammering onto a formwork wall knob (WALL). In the field, participants performed striking tasks on two surface types (a scaffold fitting/knob and wood). Across both settings, participants completed the tasks using four scaffold hammers: Estwing 16 oz (H16), Estwing 20 oz (H20), Estwing 22 oz (H22), and Stiletto 15 oz (HS).

Primary outcomes included upper-extremity muscle activity, joint kinematics (range of motion; ROM), and applied/impact forces. ROM was captured using wearable inertial measurement units (IMUs) placed on multiple body segments (head, torso, arms, and hands) to quantify movement patterns during repeated strikes. Muscle activity was measured using surface electromyography (EMG) sensors placed over key upper-limb muscles, including the anterior deltoid, middle deltoid, posterior deltoid, biceps, and a forearm muscle group associated with gripping/strike control. For field trials only, tri-axial accelerometers were additionally mounted to the hammer head and the participant's forearm to characterize tool/arm acceleration and impact forces during striking.

Applied and impact forces were recorded during striking trials to quantify force demands at the tool-surface interface. Self-reported exertion ratings were collected to capture perceived effort during each condition. In both laboratory and field settings, sensor data were time-synchronized to the task protocol and observational records to support segmentation of active striking periods and condition-specific comparisons. Data were processed and analyzed using standard biomechanical and signal-processing procedures appropriate for IMU, EMG, force, and accelerometer signals.

The Hammer Design research initiative was approved by Conestoga College's Research Ethics Board (REB #620). This project was conducted in accordance with the Declaration of Helsinki. Participants' informed consent was obtained prior to participation, including permission for publication of censored photographs and images included herein.

Code	Description	Role
HS	Standard hammer	Reference / control
H16	16 oz head weight	Lighter alternative
H20	20 oz head weight	Medium weight
H22	22 oz head weight	Heavier alternative

Figure 1 Hammer conditions in both field and laboratory studies



8 Findings

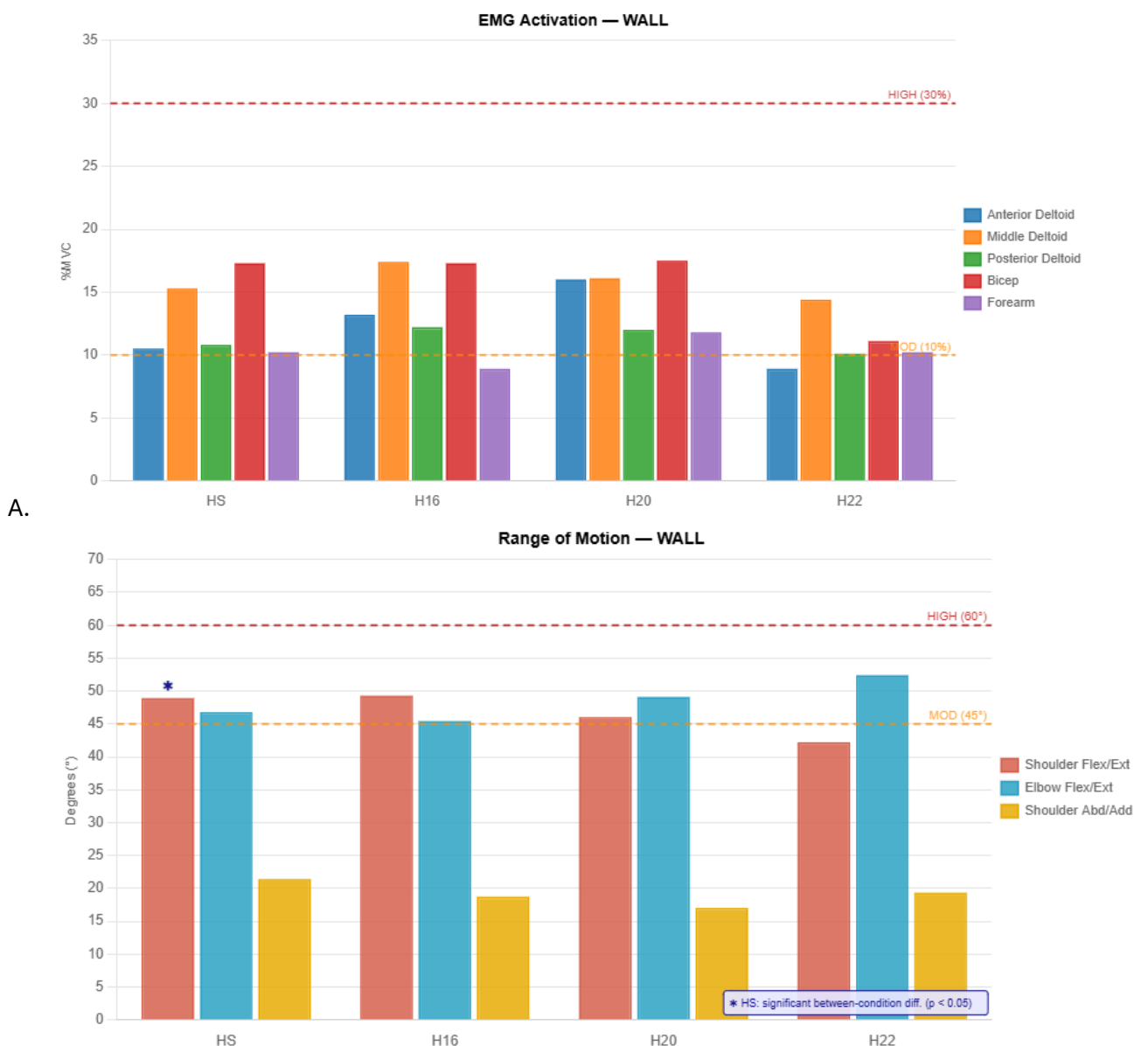
8.1 Laboratory

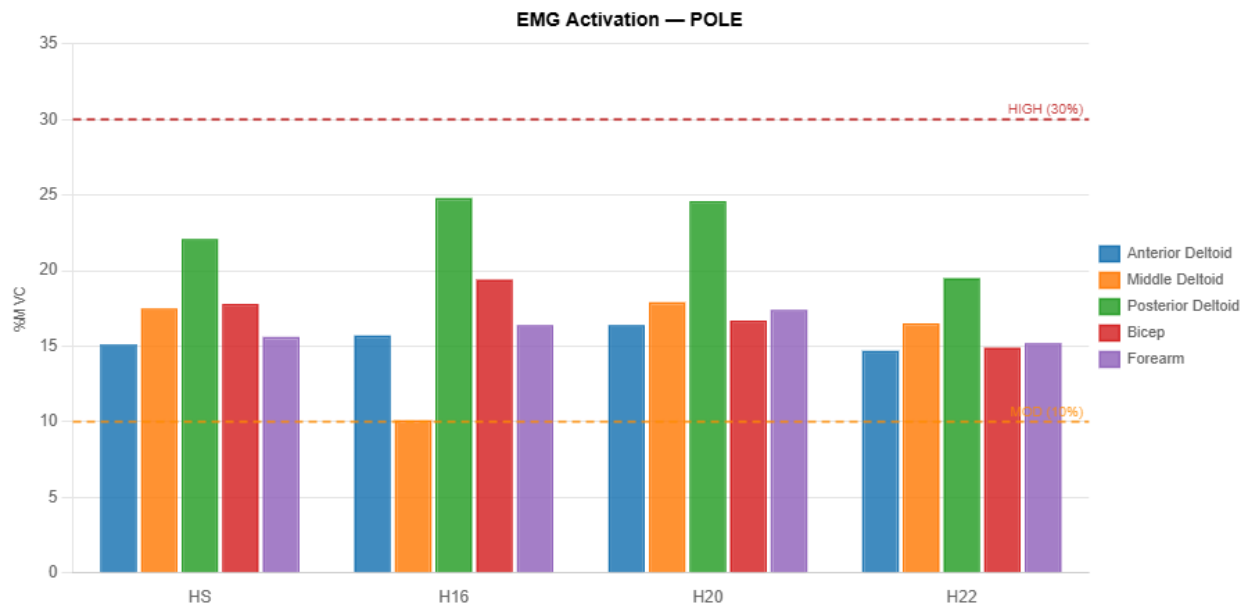
All three laboratory task conditions (WALL, FLOOR, POLE) were performed with all four hammers, resulting in complete observations across conditions. For the purposes of the risk analysis, floor-level striking was prioritized for closer review because it consistently produced the largest shoulder flexion ranges and higher posterior deltoid demands relative to the other laboratory conditions.



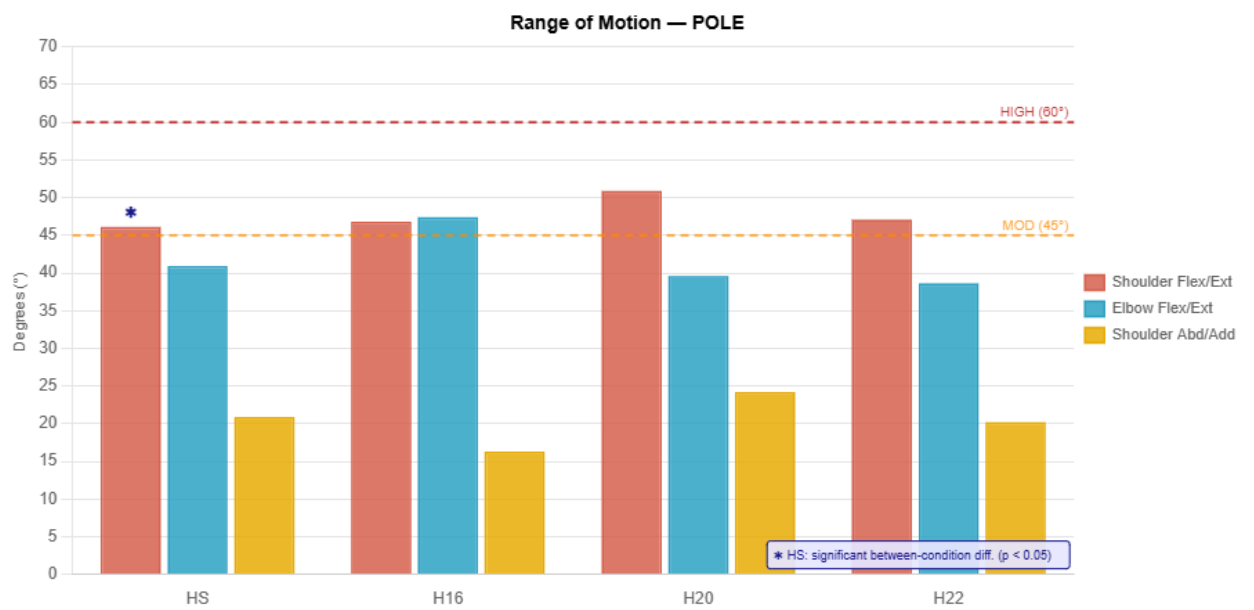
Key observations:

- **Shoulder flexion ROM:** Mean shoulder flexion ROM (ShFLEXT) during FLOOR exceeded the high-risk ROM threshold ($\geq 60^\circ$) for all four hammers (mean range: $64.25\text{--}74.40^\circ$).
- **Between-condition comparisons:** Shoulder flexion ROM differed by condition for the standard hammer (HS) (between-condition ANOVA: peak $p = 0.032$; mean $p = 0.033$). Similar condition-driven patterns were observed for the other hammers but did not reach statistical significance in this sample.
- **Posterior deltoid demand:** Mean posterior deltoid (PDEL) activation during FLOOR was higher than mean muscle activation during WALL or POLE for H16 ($31.50\% \text{MVC}$) and H20 ($31.63\% \text{MVC}$), suggesting greater deceleration demand in the floor-striking posture.





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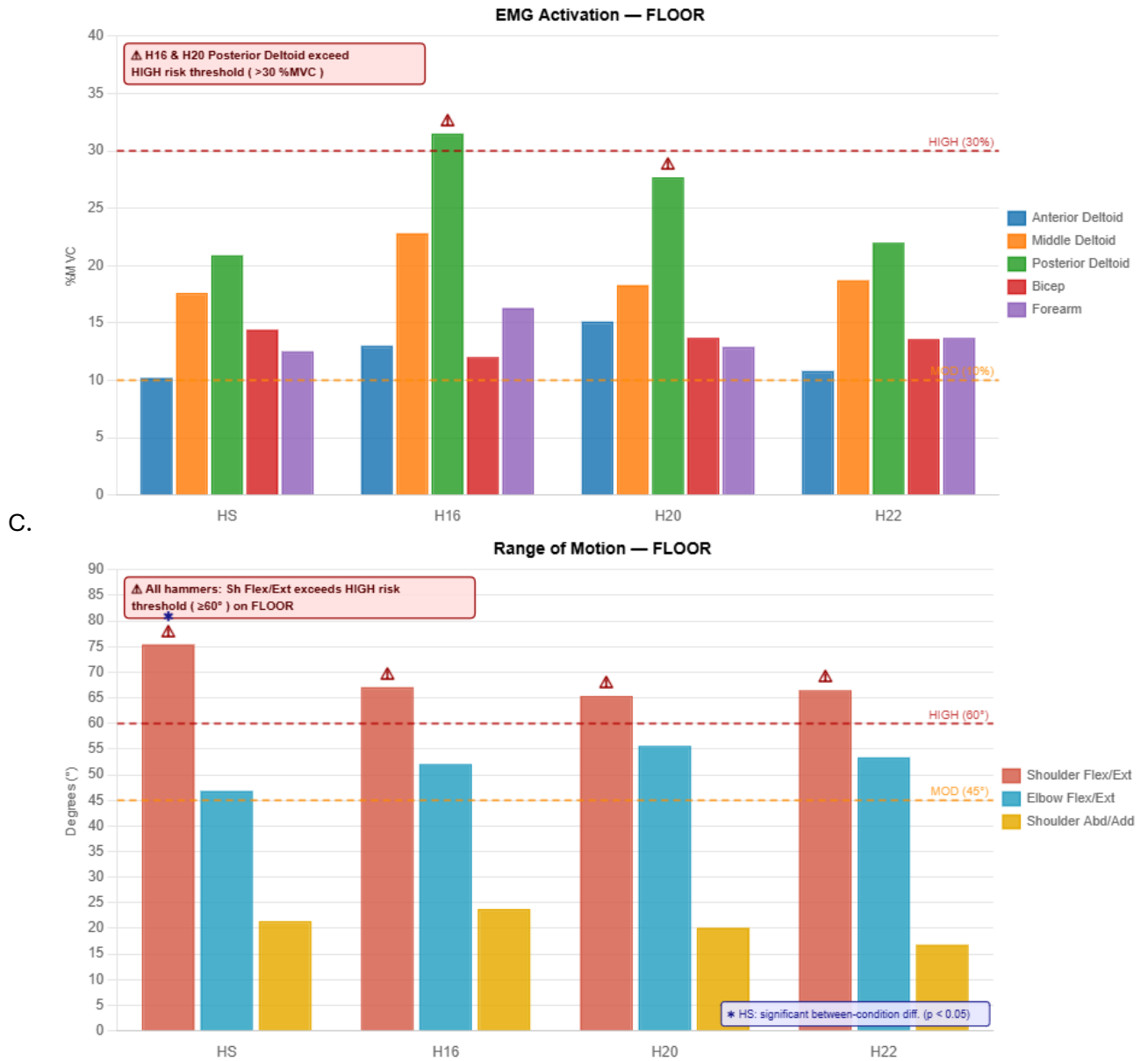
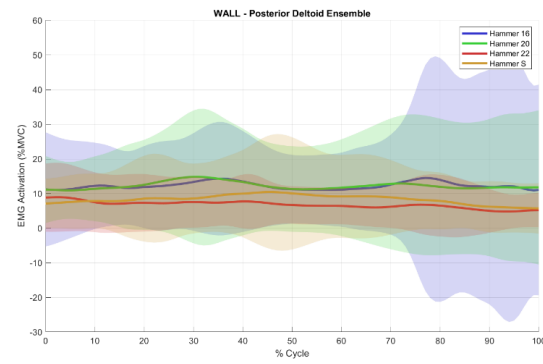
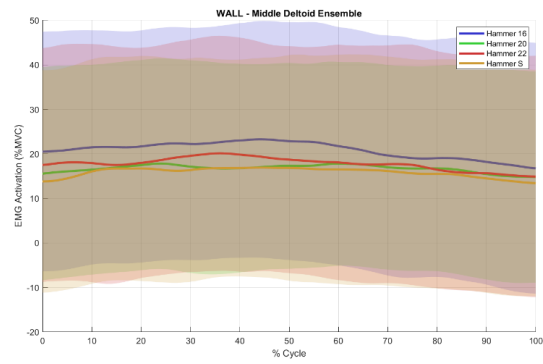
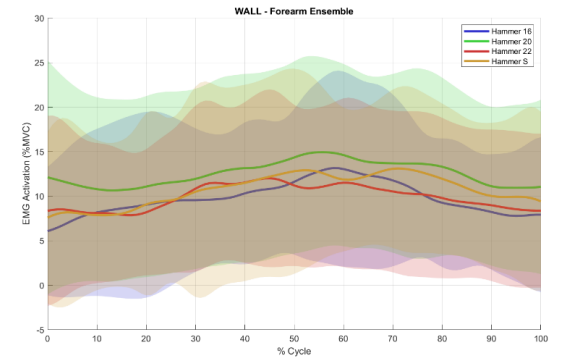
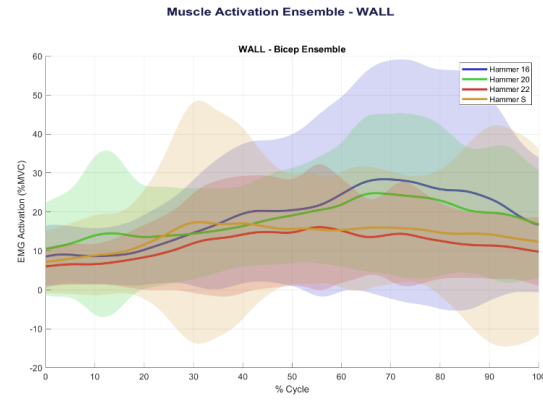
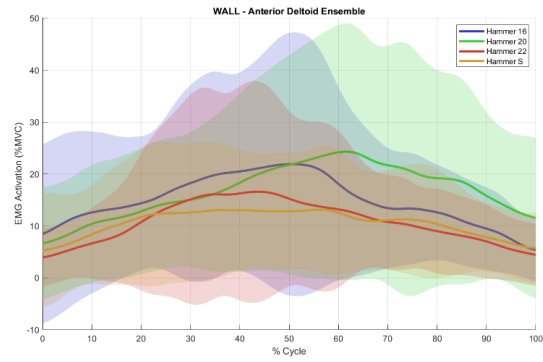


Figure 2 Condition risk summary plots, hammer comparison for range of motion and muscle activation. (A). WALL condition. (B) POLE condition. (C) FLOOR condition.



Hammer Design Report

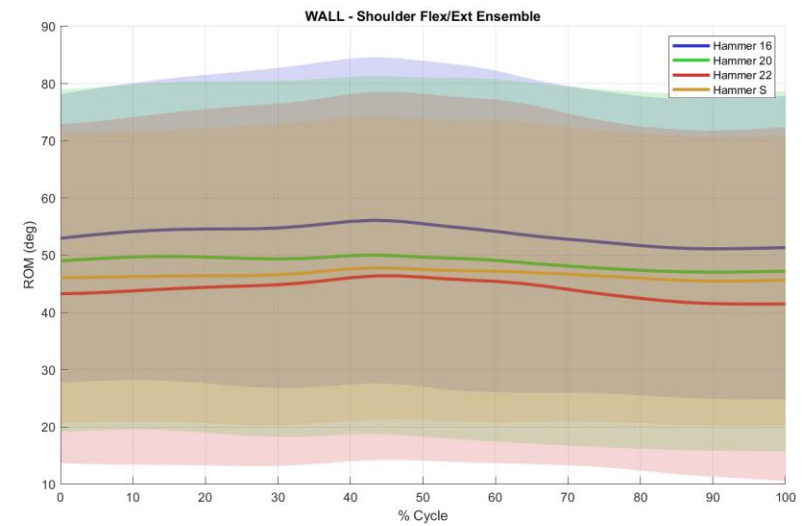
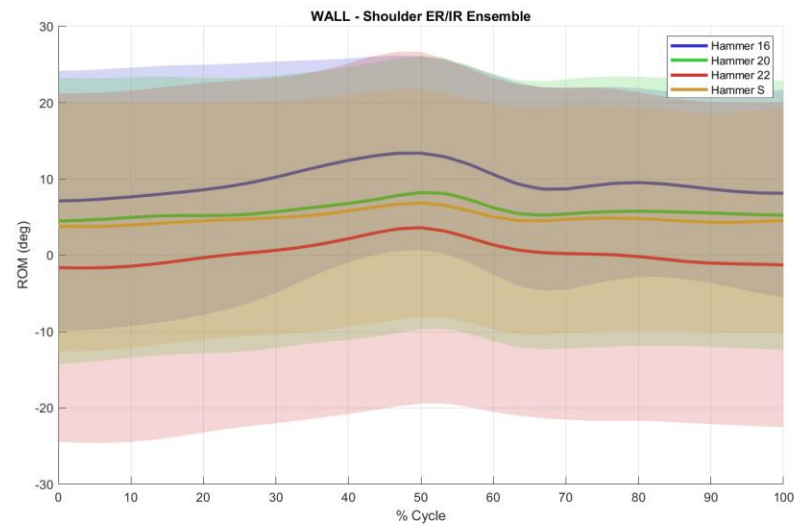
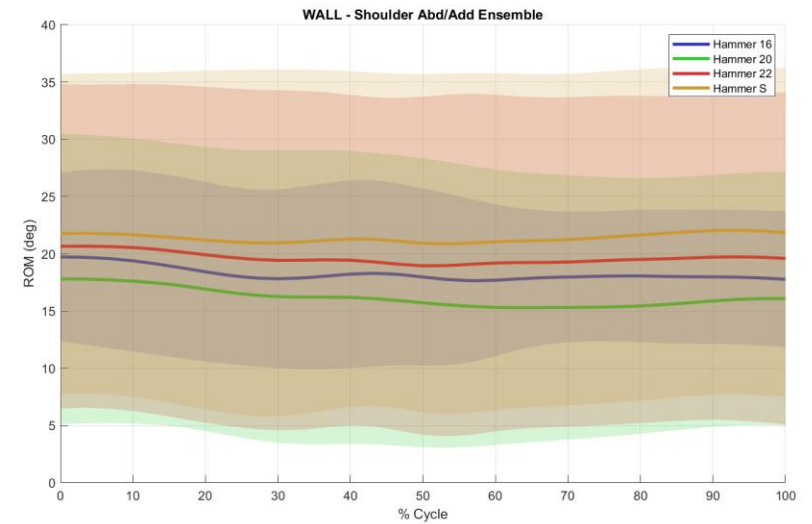
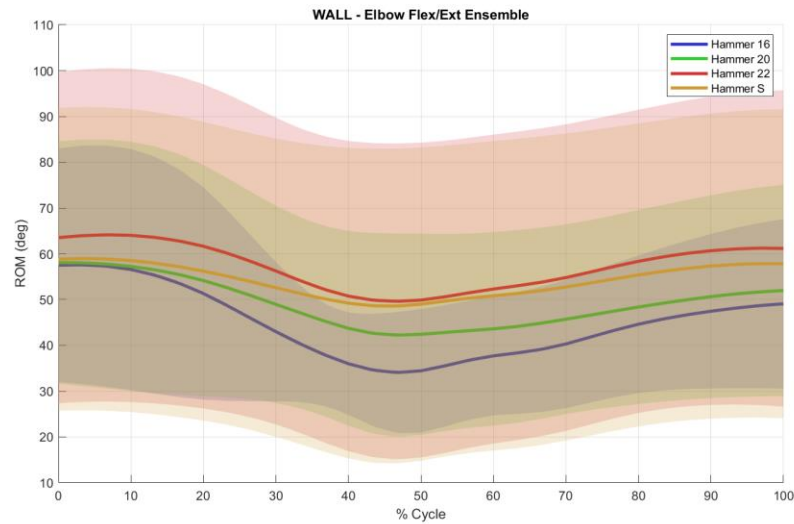
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Hammer Design Report

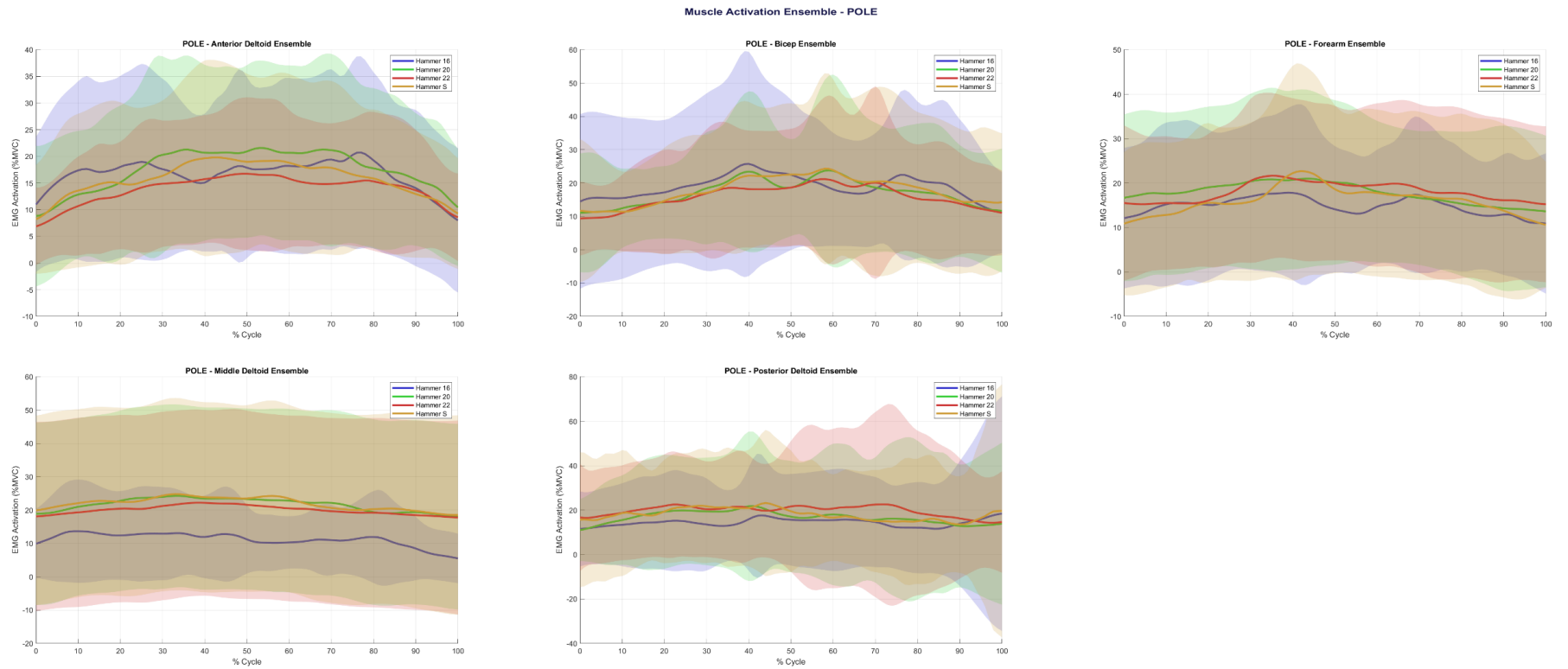
Range of Motion Ensemble - WALL





Hammer Design Report

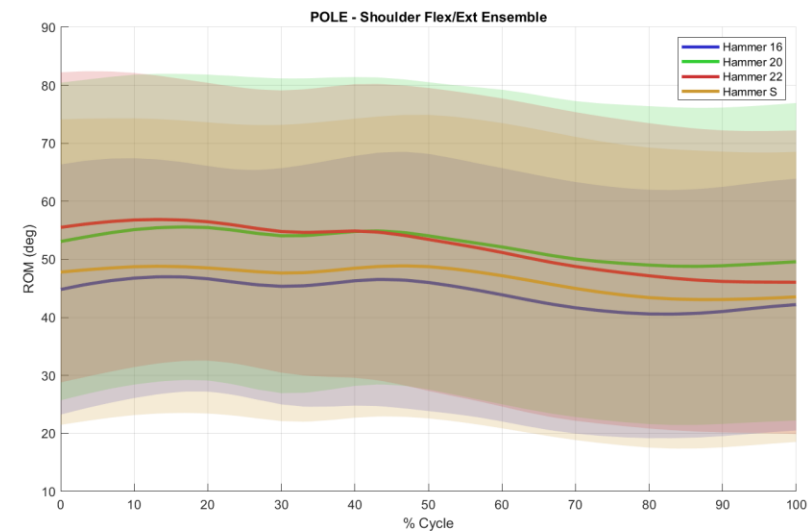
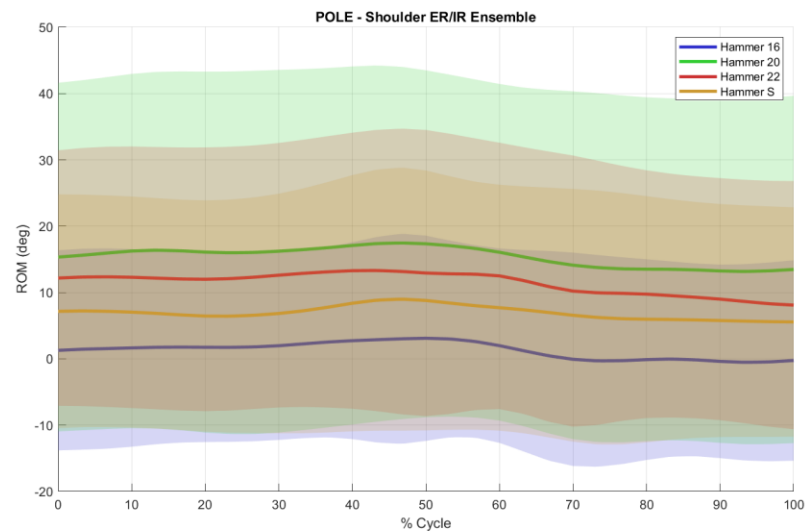
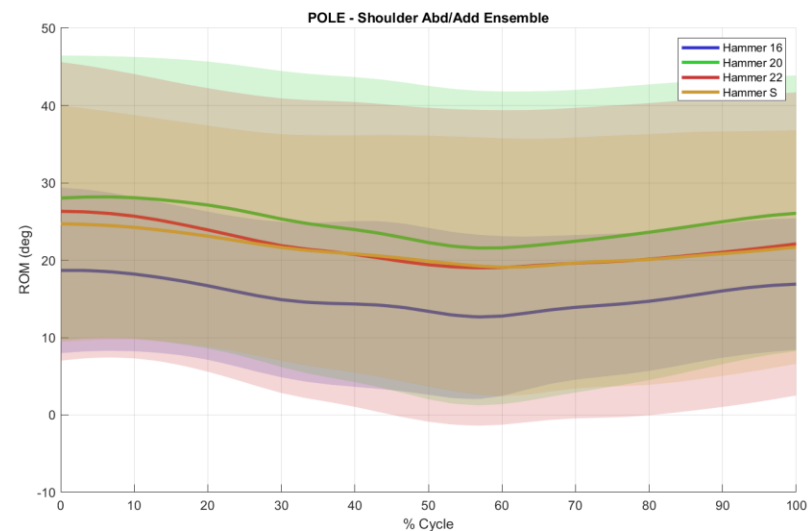
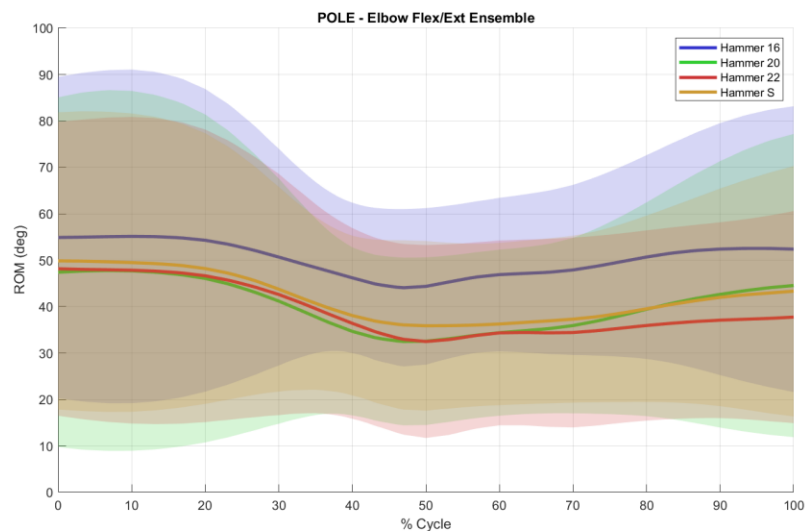
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Hammer Design Report

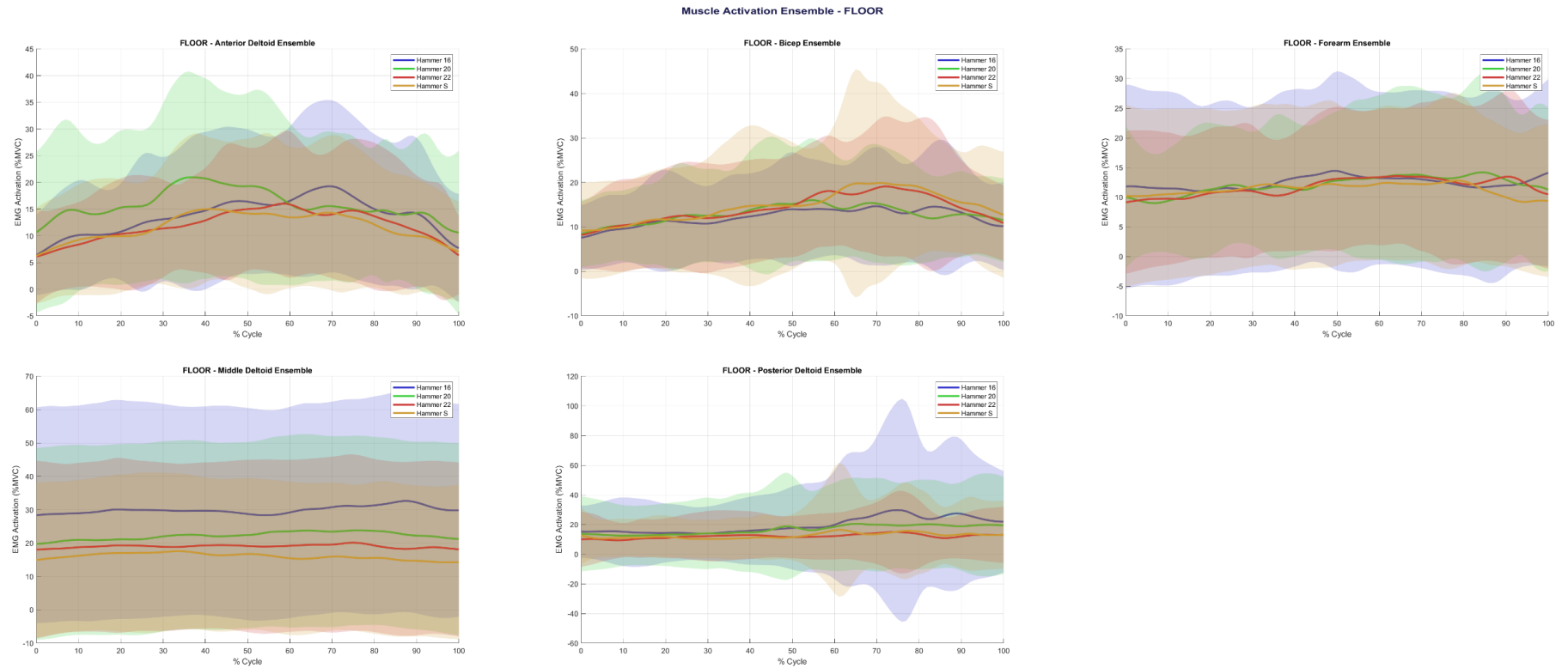
Range of Motion Ensemble - POLE





Hammer Design Report

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Hammer Design Report

Range of Motion Ensemble - FLOOR

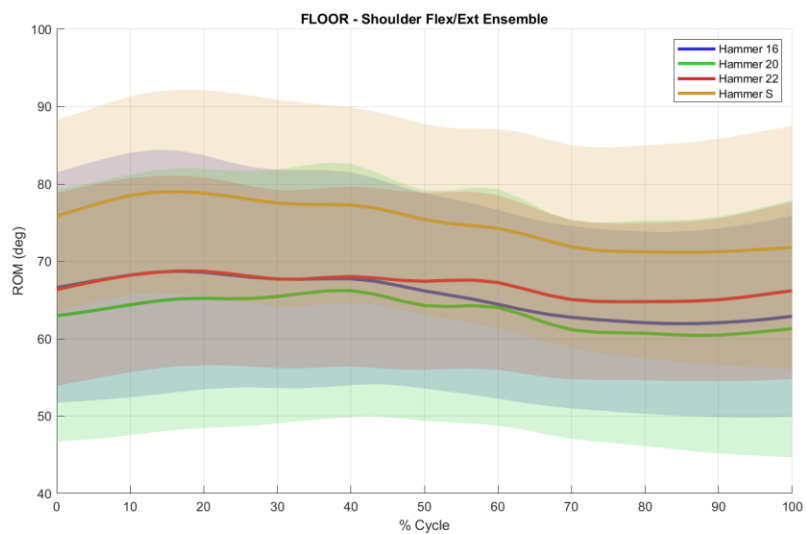
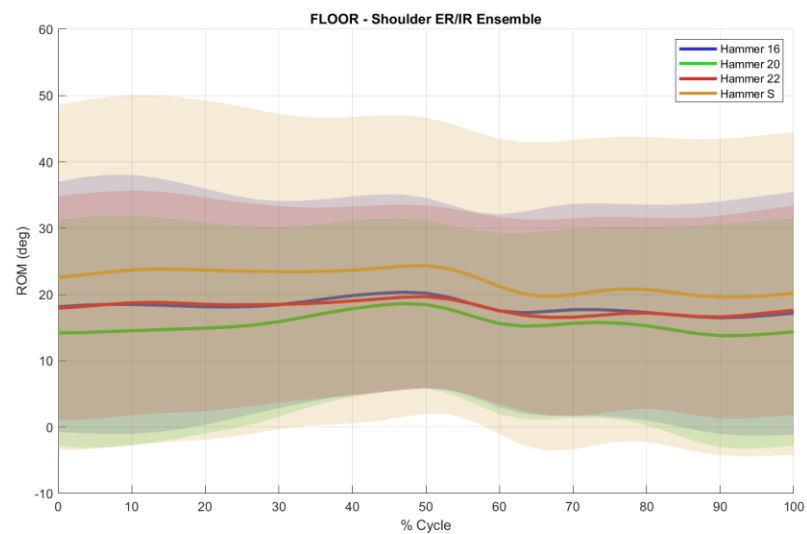
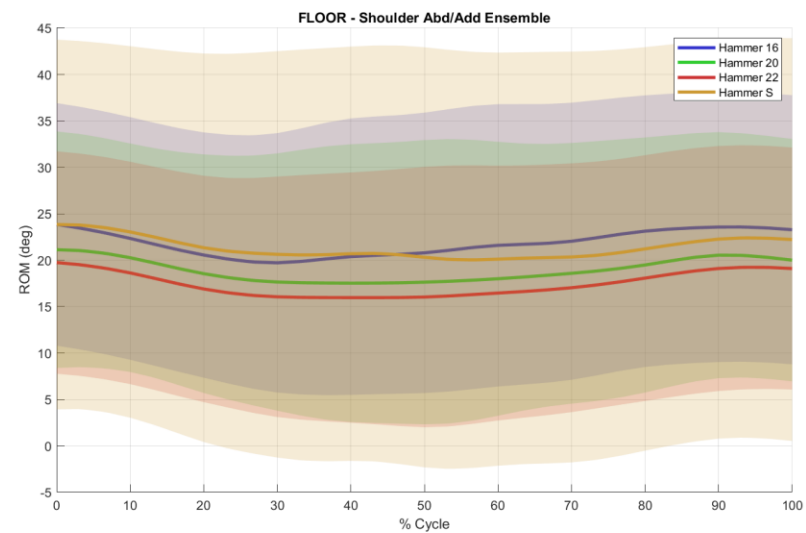
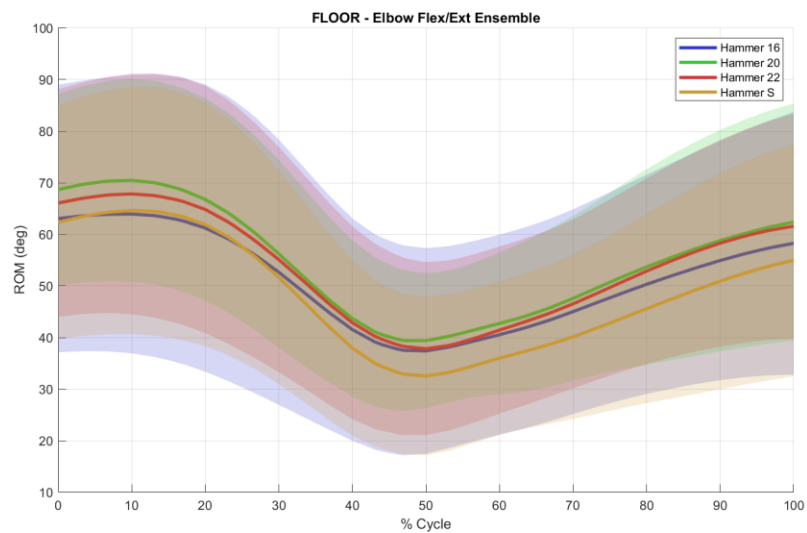




Figure 3 Ensemble plots for muscle activation and range of motion across hammers. (A). WALL condition. (B) POLE condition. (C) FLOOR condition.



Effectiveness and Efficiency scores:

Effectiveness score (how much joint movement per one unit increase in muscle exertion):

Calculation: Mean peak ROM across 4 joints / mean peak EMG across 5 muscles

Interpretation: higher score = more joint mobility per unit of muscle effort (i.e., reduced muscle effort for the achieved joint motion)

Note: In a hammering task, effectiveness captures how efficiently a worker delivers the required strike motion (joint ROM) for a given level of muscular effort (EMG). A higher score indicates the tool enables the necessary striking motion with proportionally less muscle activation — an ergonomically desirable outcome reflecting reduced muscle effort for the work performed.

Strike efficiency (how many hammer strikes per one unit increase in muscle exertion):

Calculation: N strikes / mean peak EMG

Interpretation: higher score = more strikes per unit activation)

Table 1 Summary of findings of Effectiveness and Efficiency evaluation on hammers across conditions for laboratory study.

Insight	Detail
H22 is the overall best design	Leads effectiveness on Wall & Pole conditions; leads strike efficiency on Wall & Floor. Consistently low muscle activation.
HS floor scores must be interpreted carefully	It ranks # 1 in effectiveness at Floor – but only because it drives High-risk shoulder flexion (75 degrees). H22 matches it with 9 degrees less shoulder elevation
H16 consistently underperforms	Ranks last in effectiveness on every condition. Not a competitive option ergonomically
H20 is last in strike efficiency everywhere	Despite being mid-weight, H20 never ranks high in strike efficiency. Workers get fewer strikes per unit effort compared to all other hammers.
Pole posture disrupts H22's edge	At pole level, HS is ranked first on strike efficiency. H22 drops to Rank 3. Posture-specific design matters – H22 is optimized for



forward/overhead striking but not for constrained-angle pole work.

Effectiveness & Strike Efficiency Rankings

Effectiveness Score = Mean peak ROM (°) ÷ Mean peak EMG (%MVC) | Higher = better

Condition	Rank 1		Rank 2		Rank 3		Rank 4	
	Hammer	Eff. Score	Hammer	Score	Hammer	Score	Hammer	Score
Effectiveness — WALL	H22	1.574	HS	1.340	H16	1.306	H20	1.242
Effectiveness — FLOOR	HS*	1.525	H22	1.517	H20	1.361	H16	1.138
Effectiveness — POLE	H22	1.175	H20	1.119	HS	0.994	H16	0.932

*HS Floor Rank 1 is accompanied by the highest shoulder flexion ROM (74.40°, HIGH risk). H22 comes within 0.008 of HS while producing 8.88° less shoulder flexion.

Strike Efficiency = N strikes ÷ Mean peak EMG (%MVC) | Higher = better

Condition	Rank 1		Rank 2		Rank 3		Rank 4	
	Hammer	Eff. Score	Hammer	Score	Hammer	Score	Hammer	Score
Strike Efficiency — WALL	H22	0.724	HS	0.671	H16	0.619	H20	0.578
Strike Efficiency — FLOOR	H22	0.688	HS	0.642	H16	0.515	H20	0.464
Strike Efficiency — POLE	HS	0.525	H16	0.521	H22	0.495	H20	0.444

Figure 4 Efficiency and Effectiveness Hammer Rankings per Condition for laboratory study

Detailed Risk by Condition — HS Standard Hammer

Metric	Wall	Risk	Floor	Risk	Pole	Risk
ADELTA Mean (%MVC)	9.50	LOW	10.18	MOD	15.32	MOD
PDELTA Mean (%MVC)	12.61	MOD	23.46	MOD	24.65	MOD
BICEP Mean (%MVC)	17.37	MOD	14.23	MOD	17.50	MOD
Sh_FL_EXT (°)	46.90	MOD	74.40	HIGH	46.20	MOD
El_FL_EXT (°)	48.36	MOD	45.76	MOD	40.25	LOW
Effectiveness Score	1.340 (Rank 2)		1.525 (Rank 1)		0.994 (Rank 3)	
Strike Efficiency	0.671 (Rank 2)		0.642 (Rank 2)		0.525 (Rank 1)	

A.

**Detailed Risk by Condition — H22 (22 oz)**

Metric	Wall	Risk	Floor	Risk	Pole	Risk
ADELT Mean (%MVC)	7.33	LOW	10.81	MOD	14.91	MOD
MDELT Mean (%MVC)	14.67	MOD	20.35	MOD	17.41	MOD
PDELT Mean (%MVC)	11.92	MOD	24.56	MOD	21.73	MOD
Sh_FL_EXT (°)	39.15	LOW	65.52	HIGH	47.00	MOD
EL_FL_EXT (°)	54.00	MOD	52.53	MOD	38.08	LOW
Effectiveness Score	1.574 (Rank 1)		1.517 (Rank 2)		1.175 (Rank 1)	
Strike Efficiency	0.724 (Rank 1)		0.688 (Rank 1)		0.495 (Rank 3)	

B.

Detailed Risk by Condition — H16 (16 oz)

Metric	Wall	Risk	Floor	Risk	Pole	Risk
ADELT Mean	13.17	MOD	12.96	MOD	15.67	MOD
MDELT Mean	17.35	MOD	22.81	MOD	10.11	MOD
PDELT Mean	12.2	MOD	31.5	HIGH	24.85	MOD
BICEP Mean	17.33	MOD	11.98	MOD	19.37	MOD

C.

Detailed Risk by Condition — H20 (20 oz)

Metric	Wall	Risk	Floor	Risk	Pole	Risk
ADELT Mean	15.98	MOD	15.08	MOD	16.37	MOD
MDELT Mean	16.14	MOD	18.32	MOD	17.92	MOD
PDELT Mean	11.98	MOD	27.67	MOD	24.6	MOD
BICEP Mean	17.51	MOD	13.66	MOD	16.72	MOD

D.



Figure 5 Detailed Risk by Condition for laboratory study. (A) HS, (B) H22, (C) H16, (D) H20.

8.2 Field

All four hammer conditions were assessed on both strike surfaces, yielding 41 participant-level records (21 knob; 20 wood), reflecting multiple trial records across hammer and surface conditions for the 10 field participants. Surface conditions were therefore approximately balanced in the dataset. Note: Comfort ratings in both laboratory and field settings are based on 9 participants due to one participant exclusion from the comfort analysis.

Risk-screening criteria: Conditions were flagged for priority review when they showed (1) higher absolute muscle activation ($\geq 30\%$ MVC), (2) shoulder flexion ROM $\geq 45^\circ$ (moderate-to-high postural exposure), and/or (3) clear between-hammer or between-surface differences. These criteria were used to guide where to focus interpretation and recommendations; they are not intended to imply that non-flagged conditions were risk-free.



Force attenuation profile:

When a hammer strikes a surface, the impact generates a force that travels through the tool handle into the worker's hand and arm. Repeated exposure to this transmitted force contributes to mechanical shock and cumulative musculoskeletal stress, both recognized occupational health risks for workers performing repetitive striking tasks.

- Peak Force (N): The maximum force recorded at the handle during a strike, measured in Newtons. Higher values mean more force is transmitted to the worker's grip at the moment of impact. Heavier hammer heads generally generate greater peak forces.
- Attenuation (%) — Knob & Wood:
 - What it is: How much impact force is absorbed between the hammer head and the worker's handle/forearm.
 - How it's calculated:

$$\text{Transmission_Ratio} = \text{Root mean square_forearm_force}^* / \text{Root mean square_hammer_force}^*$$

$$\text{Attenuation_}\% = (1 - \text{Transmission_Ratio}) \times 100$$

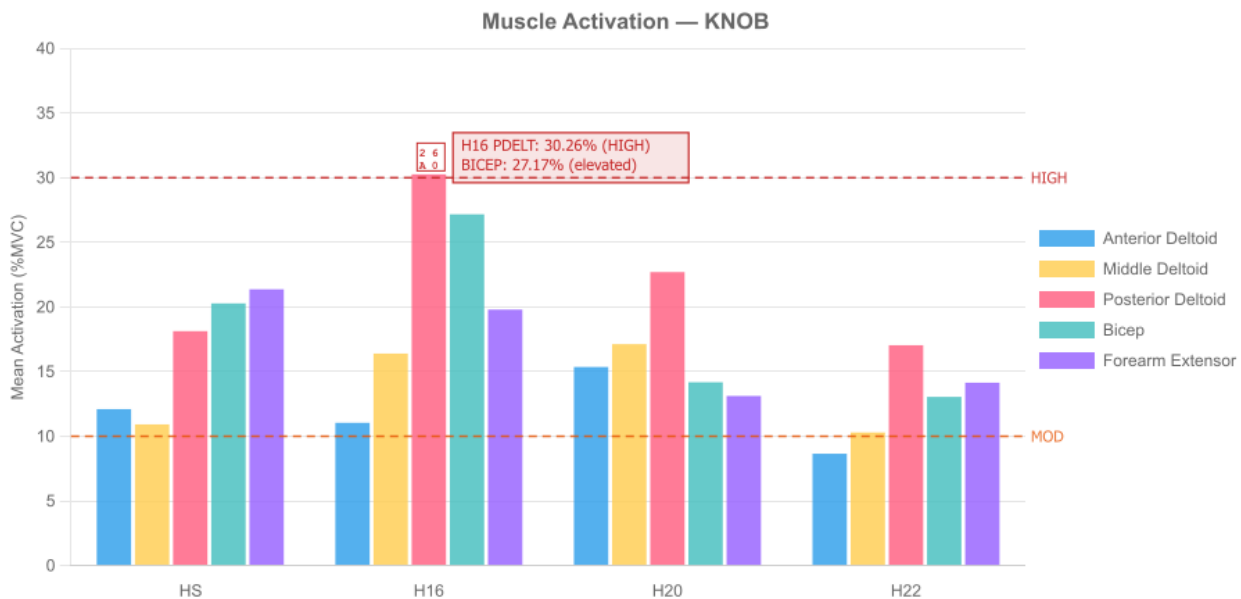
- How it's interpreted:
 - Positive % = force is attenuated (damped) — desirable.
 - Negative % = force amplification — the handle receives more force than the head delivers. *This is an impact force concern.*
- *Measured via accelerometry attached to the forearm and hammer head.*

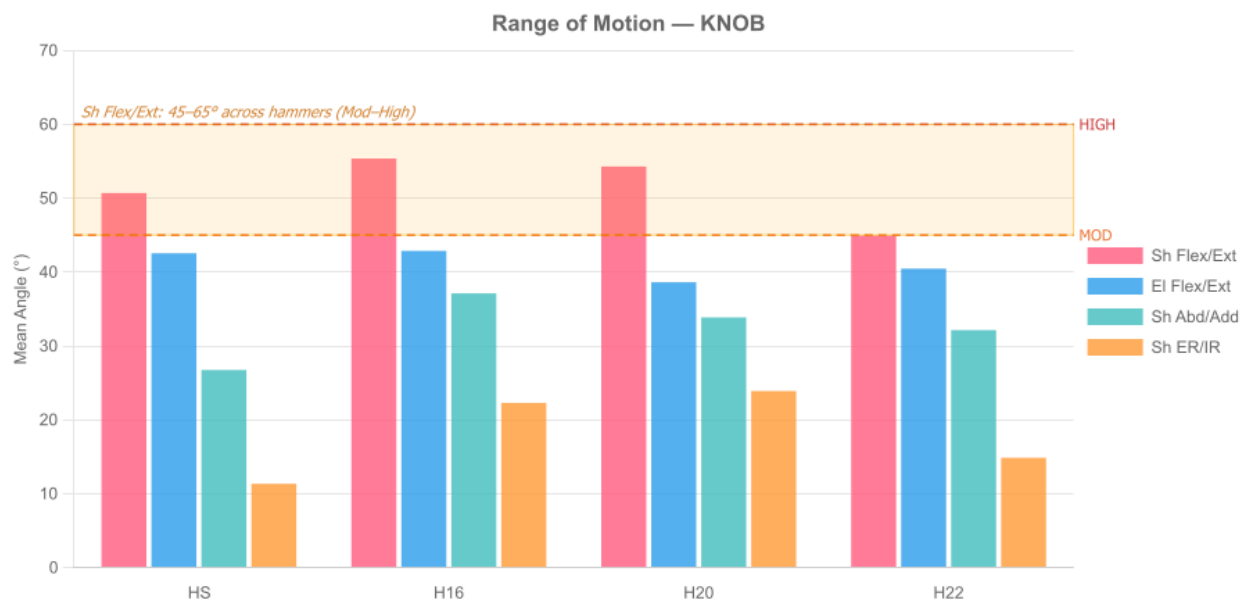


Key observations:

- **HS on wood:** Highest forearm extensor activation (mean 32.8% MVC) and statistically higher than the other hammer conditions (omnibus group comparison, $p = 0.026$).
- **H16 on knob:** Elevated posterior deltoid (PDELT) activation (45.12% peak; 30.26% mean) and higher biceps activation (42.35% peak; 27.17% mean) relative to other conditions.
- **Knob surface (all hammers):** Shoulder flexion ROM consistently in the moderate-to-high range (approximately 51–65°).
- **H20 and H22 on knob:** Higher peak impact forces than HS and H16 ($p < 0.0001$), indicating greater delivered impact on the harder fitting/knob surface.
- **H16 shows -2.6% (Knob) / -2.5% (Wood)** — force amplification, making it least suitable for hard-surface tasks.

A.





B.

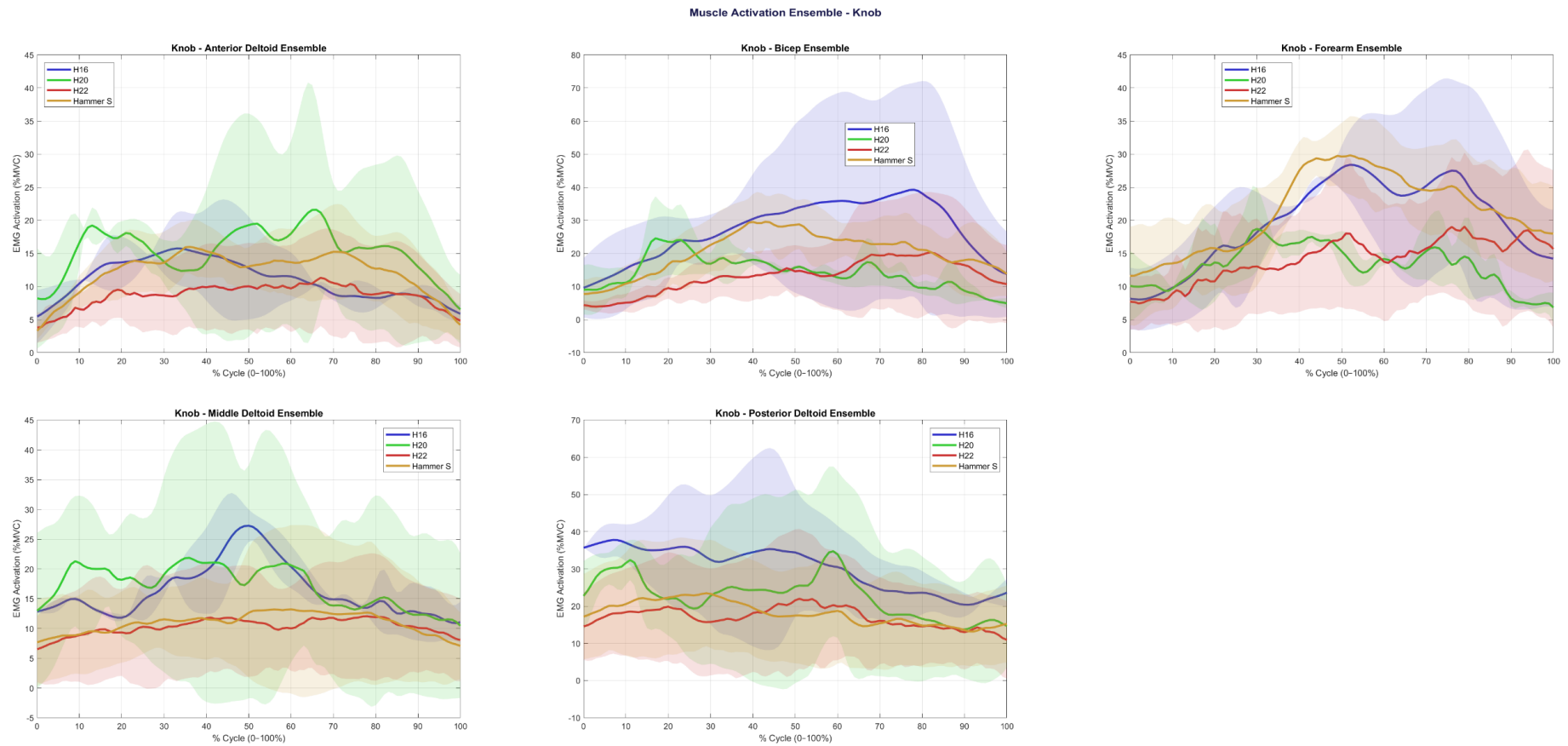


Figure 6 Condition risk summary plots, hammer comparison for range of motion and muscle activation. (A). KNOB condition. (B) WOOD condition.



Hammer Design Report

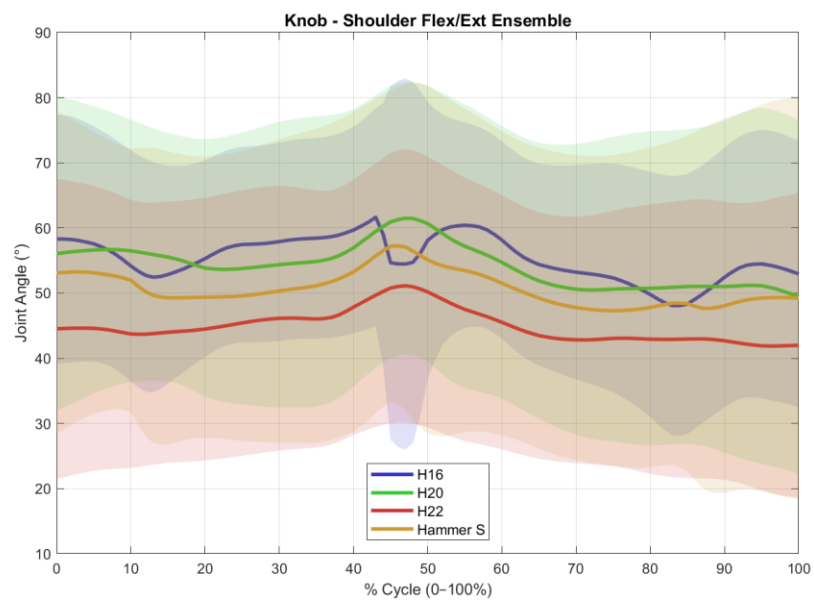
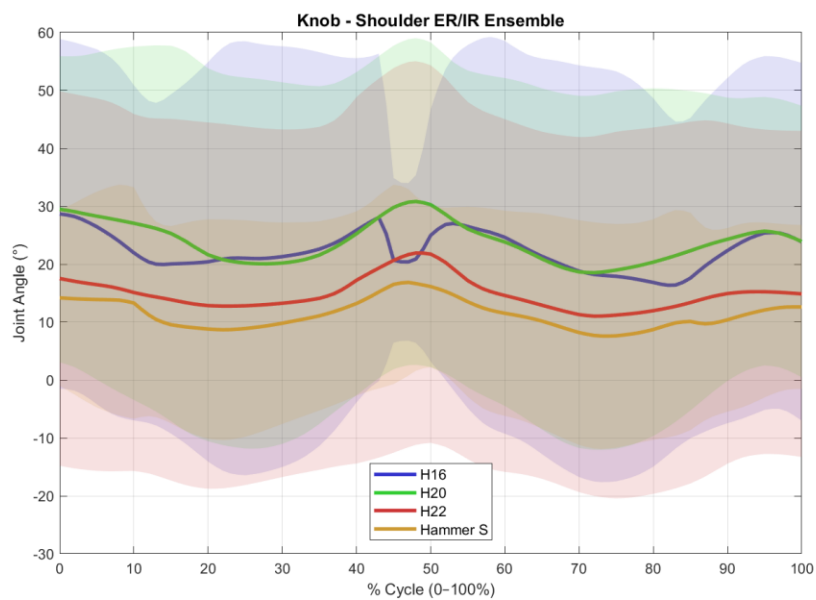
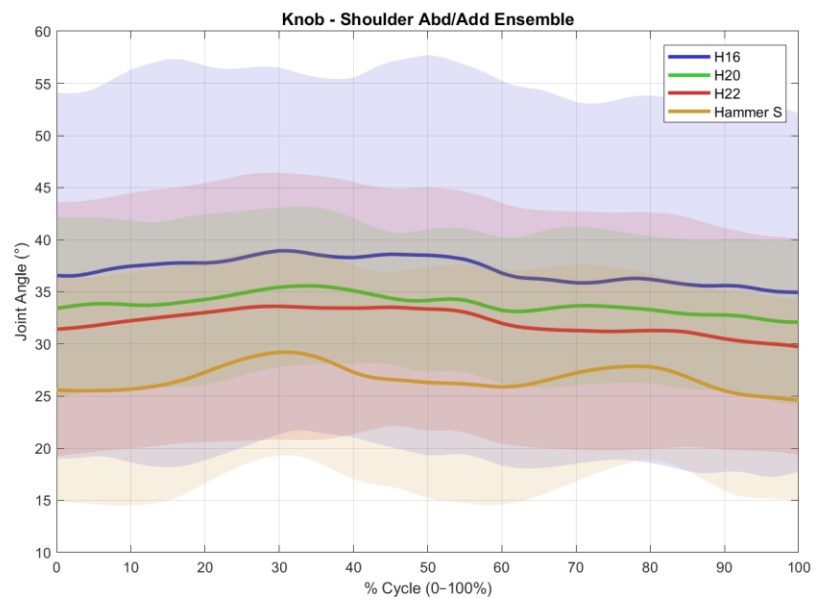
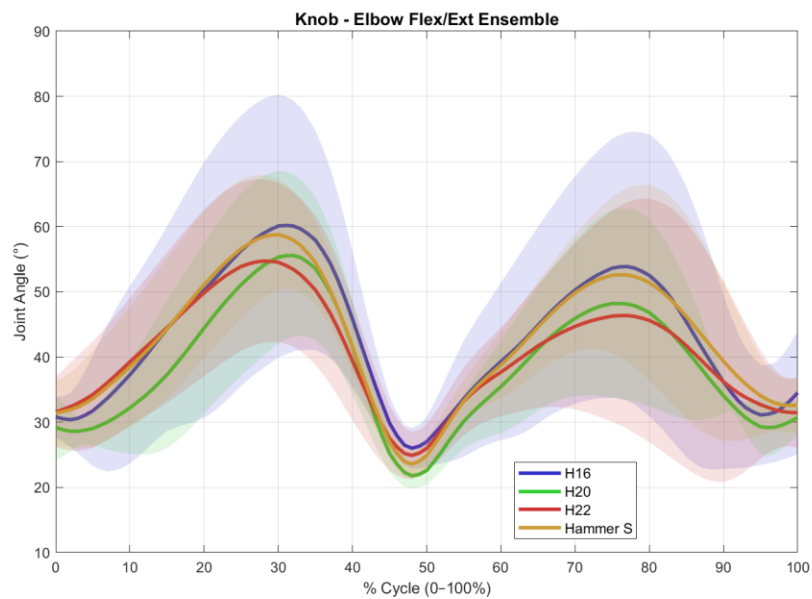
A.





Hammer Design Report

Range of Motion Ensemble - Knob

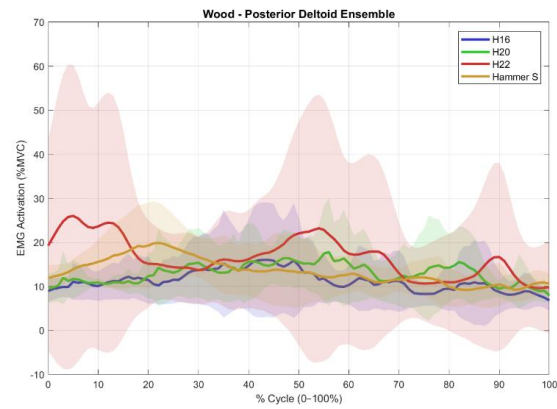
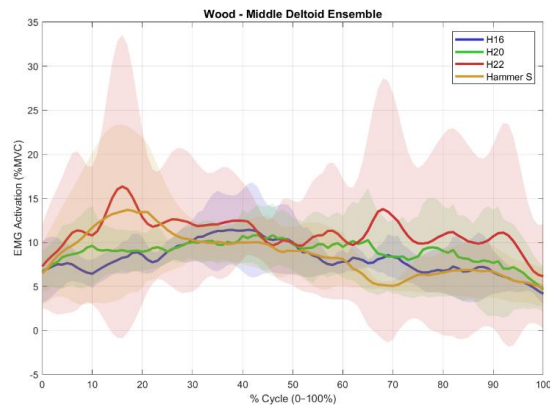
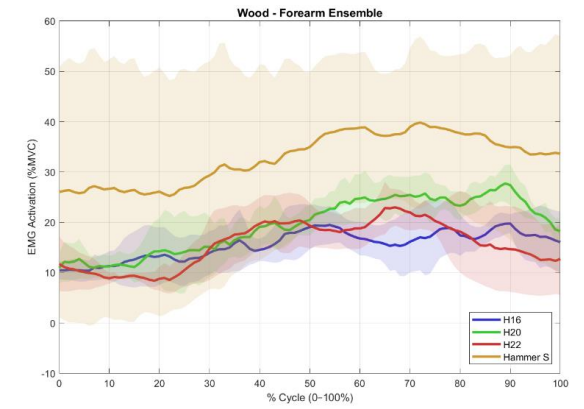
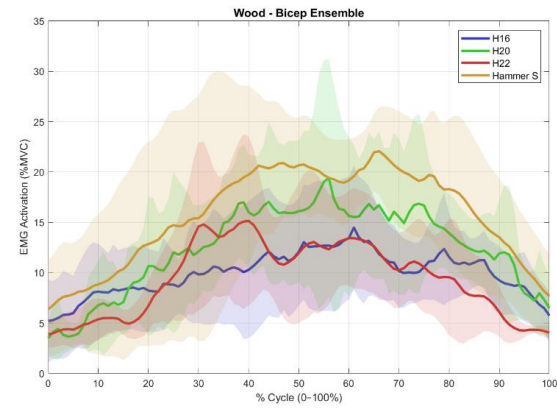
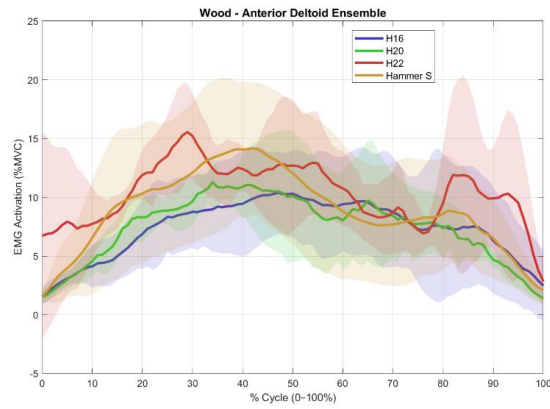




Hammer Design Report

B.

Muscle Activation Ensemble - Wood





Hammer Design Report

Range of Motion Ensemble - Wood

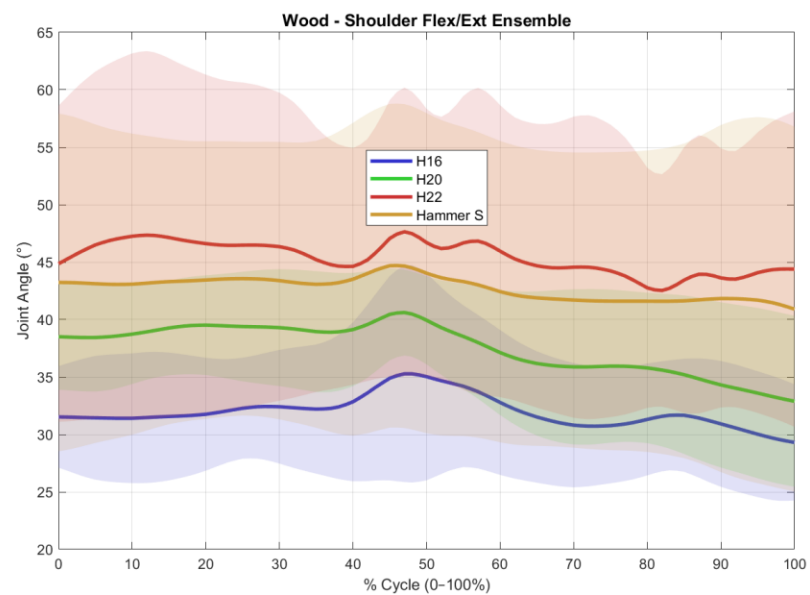
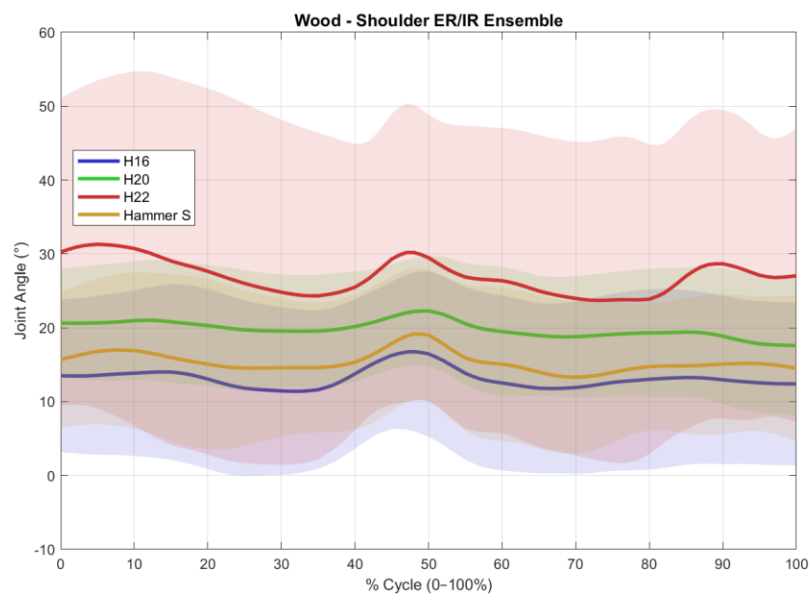
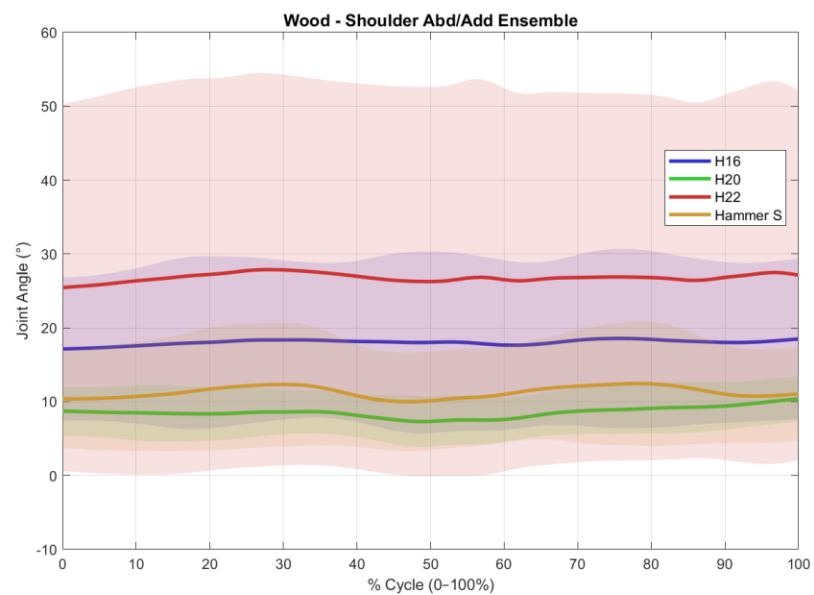
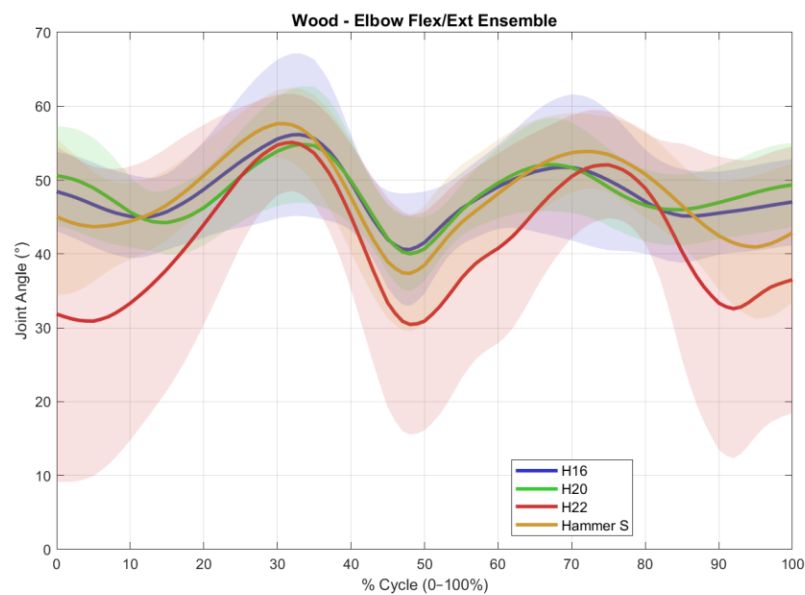




Figure 7 Ensemble plots for muscle activation and range of motion across hammers. (A). KNOB condition. (B) WOOD condition.





Effectiveness and Efficiency scores:

Table 2 Summary of findings of Effectiveness and Efficiency evaluation on hammers across conditions for field study.

Insight	Detail
H22 leads effectiveness on both surfaces — by a wide margin over HS	H22 ranks #1 for effectiveness on both Knob (2.62) and Wood (2.78)—the only hammer to lead on both. Versus HS, it is +32% on Knob and +51% on Wood; versus H16, it is +39% on Knob. Overall, H22 delivers the highest ROM output per effort across surfaces. Note: On Wood, H16 nearly matches H22 (2.66 vs 2.78; $\Delta = +4.5\%$). H22's advantage is largest on Knob.
HS is the least effective and costliest hammer in the field	HS ranks last on Wood effectiveness (1.84; Rank 4) and near-last on Knob (1.98; Rank 3). It also has the highest EMG demand on Wood (15.89% MVC), making it the poorest output-to-effort option. - Forearm extensor activation on Wood: 32.8% MVC mean—higher than all other hammers ($p = 0.026$), indicating added grip stabilization on the smooth surface. - The standard hammer requires high muscle effort with low ROM return—an ergonomic liability for repetitive work.
H16 shows the largest surface-driven reversal of any hammer	H16 shows the biggest surface reversal: worst EMG efficiency on Knob (20.93% MVC; Rank 4) but best on Wood (10.46% MVC; Rank 1). - Knob: higher stabilization demand (posterior deltoid peak 45.12% MVC; bicep 42.35% peak / 27.17% mean). - Wood: H16 edges H22 (12.64%) and H20 (12.25%) on EMG efficiency. Note: Specify the target surface—H16 is highest-risk on Knob but most efficient on Wood.
H22 EMG efficiency edge is clearest on Knob	On Knob, H22 is most EMG-efficient (12.63% MVC; Rank 1): 3.93 pp less effort than HS and 8.30 pp less than H16. On Wood, the gap narrows—H16 (10.46%) and H20 (12.25%) slightly beat H22 (12.64%). Note: If surfaces are unknown/mixed, H22 is the safest single choice—top (or near-top) EMG efficiency while leading effectiveness on both surfaces.
H20 is a consistent mid-pack performer — no clear advantage	H20 is consistently mid-pack (Rank 2–3) with no clear advantage: - Effectiveness (Knob): Rank 2 (2.28) - Effectiveness (Wood): Rank 3 (2.33)



	• EMG efficiency (Knob): Rank 2 (16.50%) • EMG efficiency (Wood): Rank 2 (12.25%) It never leads and never drops to last. Versus H22 on Knob, it requires 3.87 pp more muscle effort (16.50% vs 12.63%).
Knob surface elevates shoulder demand for all hammers	On Knob, shoulder flexion ROM is consistently moderate-to-high (~51–65°) across all hammers, approaching/entering the HIGH-risk threshold (≥60°). This is largely surface-driven—knob strikes require a more elevated arm posture regardless of hammer choice. Note: Surface access and reach distance are key ergonomic risk factors alongside hammer selection.

Effectiveness & EMG Efficiency Rankings

Effectiveness Score = Mean peak ROM (°) ÷ Mean peak EMG (%MVC) | Higher = better

Surface	Rank 1		Rank 2		Rank 3		Rank 4	
	Hammer	Eff. Score	Hammer	Score	Hammer	Score	Hammer	Score
Effectiveness — Knob	H22	2.62	H20	2.28	HS	1.98	H16	1.88
Effectiveness — Wood	H22	2.78	H16	2.66	H20	2.33	HS	1.84

EMG Efficiency = Mean muscle activation (%MVC) across 5 muscles | Lower = more efficient (Rank 1 = least effort)

Surface	Rank 1		Rank 2		Rank 3		Rank 4	
	Hammer	Mean EMG	Hammer	Mean EMG	Hammer	Mean EMG	Hammer	Mean EMG
EMG Efficiency — Knob	H22	12.63%	H20	16.50%	HS	16.56%	H16	20.93%
EMG Efficiency — Wood	H16	10.46%	H20	12.25%	H22	12.64%	HS	15.89%



Figure 8 Efficiency and Effectiveness Hammer Rankings per Condition for field study

Detailed Condition Risk — HS Standard Hammer

Metric	Knob	Risk	Wood	Risk
ADELT Mean (%MVC)	12.10	MOD	8.87	LOW
MDELT Mean (%MVC)	10.92	MOD	8.64	LOW
PDELT Mean (%MVC)	18.13	MOD	13.40	MOD
BICEP Mean (%MVC)	20.28	MOD	15.70	MOD
FOREARM Mean (%MVC)	21.37	MOD	32.82	HIGH
Sh_FL_EXT Mean (°)	50.70	MOD	42.75	LOW
El_FL_EXT Mean (°)	42.55	LOW	47.68	MOD

Key finding: FOREARM on Wood = 32.82% MVC (HIGH, $p = 0.026$ vs. all other hammers).

A.

**Detailed Condition Risk — H22 (22 oz)**

Metric	Knob	Risk	Wood	Risk
ADELTA Mean (%MVC)	8.65	LOW	10.33	MOD
MDELTA Mean (%MVC)	10.30	MOD	11.11	MOD
PDELTA Mean (%MVC)	17.04	MOD	16.72	MOD
BICEP Mean (%MVC)	13.05	MOD	9.40	LOW
FOREARM Mean (%MVC)	14.14	MOD	15.66	MOD
Sh_FL_EXT Mean (°)	44.95	LOW	45.43	MOD
El_FL_EXT Mean (°)	40.46	LOW	41.47	LOW

Best ergonomic profile: No HIGH-risk exceedances on either surface. Lowest overall EMG demand on Knob (12.63% MVC, Rank 1).

B.

**Detailed Condition Risk — H16 (16 oz)**

Metric	Knob	Risk	Wood	Risk
ADELTA Mean (%MVC)	11.04	MOD	7.33	LOW
MDELTA Mean (%MVC)	16.40	MOD	8.15	LOW
PDELTA Mean (%MVC)	30.26	HIGH	11.29	MOD
BICEP Mean (%MVC)	27.17	MOD	9.97	LOW
FOREARM Mean (%MVC)	19.80	MOD	15.58	MOD
Sh_FL_EXT Mean (°)	55.36	MOD	32.00	LOW
El_FL_EXT Mean (°)	42.87	LOW	48.13	MOD

Key finding: PDELTA on Knob = 30.26% MVC (HIGH). Largest surface-dependent reversal of any hammer.

C.

**Detailed Condition Risk — H20 (20 oz)**

Metric	Knob	Risk	Wood	Risk
ADELTA Mean (%MVC)	15.36	MOD	7.43	LOW
MDELTA Mean (%MVC)	17.13	MOD	8.96	LOW
PDELTA Mean (%MVC)	22.70	MOD	12.99	MOD
BICEP Mean (%MVC)	14.18	MOD	12.41	MOD
FOREARM Mean (%MVC)	13.12	MOD	19.46	MOD
Sh_FL_EXT Mean (°)	54.28	MOD	37.55	LOW
El_FL_EXT Mean (°)	38.62	LOW	48.15	MOD

Profile: All metrics within LOW–MOD range on both surfaces. No HIGH-risk exceedances.

Figure 9 Detailed Risk by Condition for field study. (A) HS, (B) H22, (C) H16, (D) H20.



Force attenuation profile

Note: Force transmission was quantified via dual-accelerometer derived attenuation ratios (forearm / hammer head). Absolute peak forces are not reported, as the 120 Hz accelerometer sampling rate is insufficient to resolve true impact transients; only relative transmission ratios are used for comparison.

- **HS- Safest force profile:** HS showed positive attenuation on Knob (+1.4%), the handle absorbed a small fraction of the impact force, reducing what the worker feels. On Wood, attenuation was near-zero (+0.1%), meaning essentially all force passed through. For workers concerned about long-term effects of impact forces, HS transmits the least mechanical shock.
- **H16- Only hammer with force amplification on both surfaces:** H16 showed negative attenuation on Knob (-2.6%) and Wood (-2.5%). This means the mechanical shock dampening at the handle in H16 actually amplified force rather than attenuating it, more energy arrives at the handle than was generated at the hammer-surface interface. Combined with elevated muscle activation on Knob, H16 presents the most concerning force-transmission profile.
- **H20- Neutral force transmission:** H20 showed near-zero attenuation on both surfaces (Knob +0.8%; Wood 0.0%), meaning force passes through largely unchanged. H20 neither attenuates nor amplifies, it is ergonomically neutral from a force-transmission standpoint.
- **H22- Highest transmitted force, slight amplification:** H22 generated the highest relative force transmission on Knob, consistent with its heavier head. Attenuation was slightly negative (-0.5% Knob; -0.3% Wood), meaning a negligible amplification effect. In practice, the magnitude of amplification is very small and unlikely to be clinically meaningful on its own, but the relative force delivered to the handle is the highest of all four designs.

Table 3 Summary interpretation for the force attenuation profile in field study across hammers

Hammer	Force Profile (relative ranking; absolute N not reported)	Attenuation	Interpretation
HS	Lowest (relative)	+1.4% / +0.1%	Best attenuation. Least absolute force at handle. Lowest effectiveness.
H16	Low-moderate (relative)	-2.6% / -2.5%	FORCE AMPLIFICATION on both surfaces. Avoid on hard/knob surfaces.
H20	Moderate (relative)	+0.8% / 0.0%	Neutral. Force passes through unchanged. No advantage or concern.
H22	Highest (relative)	-0.5% / -0.3%	Highest relative force transmission but minimal



amplification. Best overall ergonomic profile.

8.3 Comfort Ratings

After completing the tasks, each participant rated the comfort of each hammer on a scale from 0 (least comfortable) to 10 (most comfortable). Comfort reflects how the hammer felt in the hand during the hammering tasks — including grip feel, strike impact, and overall handling. These ratings capture participants' subjective experience alongside the objective biomechanical measures.



Laboratory Study

In the laboratory (9 participants, 3 conditions: Pole, Floor, Wall), comfort ratings varied modestly across hammers. The Stiletto HS received the highest overall rating (7.1 ± 2.2), particularly in the Pole condition (7.9 ± 1.8). The three Estwing hammers scored similarly, with H16 and H20 receiving slightly higher ratings than H22.

Table 4 Mean comfort ratings (0–10 scale) by hammer and condition, laboratory study (n = 9 participants per cell)

Hammer	Pole (mean $\pm$ SD)	Floor (mean $\pm$ SD)	Wall (mean $\pm$ SD)	Overall (mean $\pm$ SD)
HS (Stiletto 15 oz)	7.9 ± 1.8	7.0 ± 2.2	6.4 ± 2.3	7.1 ± 2.2
H16 (Estwing 16 oz)	6.1 ± 2.2	7.1 ± 2.0	6.8 ± 2.6	6.7 ± 2.3
H20 (Estwing 20 oz)	6.4 ± 1.8	6.8 ± 1.8	6.7 ± 2.1	6.6 ± 1.9
H22 (Estwing 22 oz)	6.1 ± 1.5	6.2 ± 2.7	6.6 ± 2.4	6.3 ± 2.2

Key takeaway: Comfort differences between hammers were small (range: 6.3–7.1 overall). No single hammer was rated poorly, and the standard Stiletto HS was most consistently preferred by participants across conditions.



Field Study

In the field (9 participants, striking on knob and wood surfaces), comfort ratings followed the same trend: HS received the highest rating (7.1 ± 2.0), followed by H16 (6.9 ± 1.6), H20 (6.3 ± 1.3), and H22 (5.9 ± 1.3). Heavier hammers were generally rated as less comfortable, likely reflecting increased fatigue and vibration during real-world conditions.

Table 5 Mean comfort ratings (0–10 scale) by hammer, field study (n = 9 participants)

Hammer	Mean Comfort Rating ($\pm$ SD)
HS (Stiletto 15 oz)	7.1 ± 2.0
H16 (Estwing 16 oz)	6.9 ± 1.6
H20 (Estwing 20 oz)	6.3 ± 1.3
H22 (Estwing 22 oz)	5.9 ± 1.3

Key takeaway: Consistent with the laboratory, the Stiletto HS was rated most comfortable in the field. H22 received the lowest comfort rating, which, combined with its higher peak force delivery, suggests increased subjective load on workers. H16 and H20 performed comparably and were rated above average by participants.

8.4 Overall Recommendations

Using the body-worn sensor findings from both the laboratory and field components of this study, the recommendations below prioritize the most practical opportunities to reduce upper-extremity loading, discomfort, and fatigue during repetitive hammering while assembling scaffolding. Overall, the evidence indicates that strike condition and task setup (e.g., target surface compliance and strike height/reach) are primary drivers of shoulder and forearm demand, with hammer selection acting as an important but secondary control that should be matched to the surface and work requirements. In addition, the findings provide evidence-informed insight into which hammer design is most preferable overall (and in some cases by surface), supporting more deliberate tool selection rather than relying on tradition or personal preference alone. Recommendations are organized according to the Hierarchy of Controls (Figure 10), progressing from preferred and most effective controls (elimination/substitution/engineering) to administrative controls and PPE, and are intended to be implemented in combination where needed to adequately protect workers while maintaining task performance.

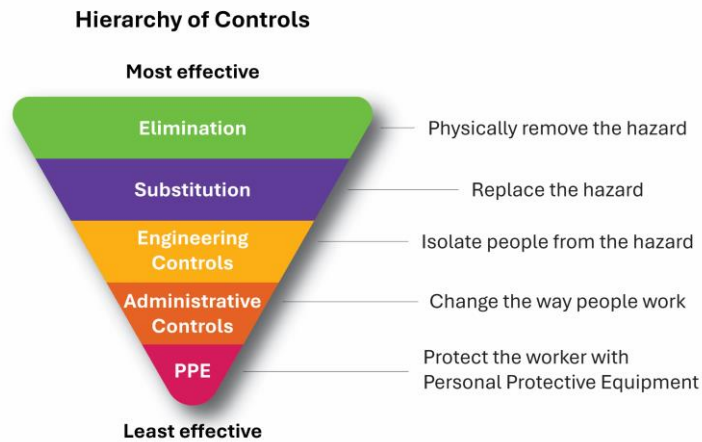


Figure 10 Hierarchy of controls framework used to organize the recommendations in this report for reducing upper extremity loading during repetitive hammering tasks performed on elevated platforms (e.g., scaffolding assembly/disassembly). The hierarchy identifies the preferred order of actions to control hazards—starting with elimination/substitution and engineering controls, followed by administrative controls and PPE. Modified from CCOHS “Hazard and Risk – Hierarchy of Controls”.



Field Study Recommendations

The following recommendations are derived from EMG, joint ROM, peak impact force, force transmission ratio, and cumulative energy findings across four hammer conditions tested on two surface types in a live field setting. The field environment reflects authentic construction workloads and worker technique, providing high external validity.

Field Study Recommendations		
Level	Control	Recommendation
SUBSTITUTION	Tool Selection	Replace HS with H22 for Knob/scaffold striking tasks. H22 offers lowest EMG demands (rank 1) and highest effectiveness on hard surfaces.
SUBSTITUTION	Tool Selection	For Wood-surface tasks, prefer H16 or H22 over HS, which shows significantly elevated forearm activation on Wood ($p = 0.026$).
ENGINEERING	Tool Design	Investigate handle designs with improved vibration damping for H16 (attenuation = -2.6% , force amplification risk).
ENGINEERING	Workstation	Design work setups that minimize overhead/forward-reach postures to reduce Sh_FL_EXT (target $<45^\circ$).
ADMINISTRATIVE	Task Rotation	Implement hammer rotation schedules to limit continuous Knob-surface striking (highest cumulative muscle load).
ADMINISTRATIVE	Work-Rest Ratio	Schedule micro-breaks every 20–30 minutes, especially on Wood surfaces (higher energy per strike than Knob).
ADMINISTRATIVE	Training	Train workers in optimal hammer grip and strike technique to minimize unnecessary forearm extensor co-contraction.
PPE	Anti-vibration Gloves	Evaluate anti-vibration gloves for Knob-surface tasks where H20 and H22 generate the highest peak forces (5.84–6.89 N).

Figure 11 Summary of field study recommendations

Substitution Recommendations (Tool Selection)

- Replace HS with H22 for Knob/Scaffold Striking Tasks [HIGH PRIORITY]

HS consistently underperformed on knob (hard-fitting) striking in the field. For knob work, prioritize H22 as the primary option, with H20 as an alternate where similar performance is required.

Supporting evidence: H22 showed lower average upper-extremity muscle demand than HS on knob striking while maintaining effective impact delivery.

Implementation: Adopt H22 for scaffold fitting/knob striking and provide a brief orientation for workers during rollout.

- Prefer H16 or H22 Over HS for Wood-Surface Tasks [MODERATE PRIORITY]



For wood-surface striking, HS showed higher forearm extensor demand than the other hammers. For sustained wood nailing/framing work, prefer H16 (lighter) or H22 (higher impact per strike) based on task needs and worker comfort.

Supporting evidence: Field results indicated a meaningful increase in forearm extensor loading with HS on wood compared with the other hammers.

Implementation: Standardize H16 or H22 for high-volume wood striking tasks and confirm the selection with worker feedback during a short pilot period.

Engineering Recommendations

- Mechanical Shock/Vibration Damping Improvements for H16 Handle [HIGH PRIORITY]

Field results suggest H16 may transmit (and in some conditions slightly amplify) impact forces rather than attenuating them, particularly on hard surfaces. Avoid using H16 as the default option for knob striking until vibration/force transmission performance can be improved.

Supporting evidence: H16 showed the least favourable force-transmission pattern in the field dataset, alongside higher shoulder/arm demands on knob work.

Implementation: If H16 remains in use, evaluate handle/anti-vibration options and limit its use on hard fitting/knob striking where feasible.

- Workstation Design to Minimise Shoulder Flexion [MODERATE PRIORITY]

Across field conditions, shoulder flexion remained in a moderate range, driven mainly by work height and reach distance. Reducing shoulder elevation requires task/setup changes in addition to hammer selection.

Key evidence: Field mean ShFLEXT: HS = 47.8° (MOD), H20 = 45.9° (MOD), H22 = 44.95° (low-moderate boundary). Reducing ShFLEXT to <45° requires postural interventions beyond hammer selection alone.

Implementation: Where possible, adjust work positioning so repetitive striking occurs closer to a neutral arm posture (reduced reach and shoulder elevation).

Administrative Recommendations

- Hammer Rotation Schedules for Knob-Surface Tasks [HIGH PRIORITY]

Knob-surface striking is higher demand than wood striking and can accelerate fatigue when performed continuously. Use task rotation to reduce prolonged exposure to high-repetition knob striking.

Implementation: Rotate workers between knob striking and lower-demand tasks to manage cumulative upper-extremity load.



- Micro-Break Schedule for Wood-Surface Tasks [MODERATE PRIORITY]

During prolonged wood-surface striking, schedule brief recovery breaks to reduce cumulative fatigue in the forearm and shoulder, even if perceived effort feels low.

Implementation: Build short breaks into longer wood-striking periods and encourage simple mobility/recovery movements as appropriate.

- Grip and Technique Training [MODERATE PRIORITY]

Forearm loading can increase when workers use a sustained “death grip” or brace through impact, particularly on smoother/compliant targets. Technique coaching can help reduce unnecessary co-contraction and improve control.

Implementation: Provide short, hands-on training focused on relaxed grip between strikes and controlled impact technique, then reinforce during periodic safety/ergonomics check-ins.

PPE Recommendations

- Anti-Vibration Gloves for H20/H22 Knob Tasks [MODERATE PRIORITY]

For sustained knob striking with heavier hammers (H20/H22), consider gloves that reduce vibration while still allowing secure grip and tool control.

Supporting evidence: Heavier hammers produced higher impact forces on knob striking in this dataset, which can increase vibration/force exposure at the hand.

Implementation: If gloves are introduced, confirm they do not compromise grip, precision, or comfort during typical work.



Laboratory Study Recommendations

Laboratory testing isolated the effects of hammer design across three common strike-height scenarios (wall, floor, and pole). Overall, task height and posture drove risk more than hammer choice, so recommendations prioritize changing the work setup first and using hammer selection as a supporting control.

Lab Study Recommendations		
Level	Control	Recommendation
SUBSTITUTION	Tool Selection	Replace HS with H22 for Wall and Pole tasks. H22 provides superior effectiveness (rank 1) and strike efficiency (rank 1 on Wall) with lower shoulder ROM across conditions.
SUBSTITUTION	Tool Selection	Use H22 over HS for Floor tasks. Despite near-equal effectiveness (1.517 vs. 1.525), HS produces 74.40° Sh_FL_EXT (HIGH risk) vs. H22's 65.52° — a clinically meaningful 8.88° reduction.
SUBSTITUTION	Task Redesign	Convert floor-level striking to wall-height tasks where feasible (elevated platforms, staged decking, angled attachments) to move Sh_FL_EXT from HIGH (>60°) to MODERATE range.
ENGINEERING	Tool Design	Investigate handle geometry for H16 to reduce elevated PDELT and FOREARM demands on Floor/Wall (longer handle or modified grip angle may reduce deceleration effort).
ENGINEERING	Workstation	Implement adjustable-height platforms for floor-level fastening tasks to reduce sustained forward trunk and shoulder flexion.
ADMINISTRATIVE	Task Rotation	Rotate workers off floor-level hammer striking every 20–30 minutes to limit cumulative high-ROM shoulder exposure.
ADMINISTRATIVE	Training	Train workers to use H22 with a controlled, momentum-based technique minimizing unnecessary BICEP and PDELT co-contraction during deceleration.
PPE	Support Bracing	Evaluate elbow support or anti-vibration gloves for extended floor-level tasks where elbow flexion consistently reaches moderate-risk zone (45–54°).

Figure 12 Summary of laboratory study recommendations

Substitution Recommendations (Tool Selection)

- Replace HS with H22 for Wall and Pole Tasks [HIGH PRIORITY]

For wall- and pole-height striking, H22 performed best overall in the laboratory rankings, with lower biomechanical cost than HS while maintaining strong task output. Standardize H22 for these common working-height tasks where feasible.

Supporting evidence: H22 ranked highest for effectiveness at wall height and showed lower shoulder/arm demands than HS in this condition.

Supporting evidence: On pole-height work, H22 remained a top performer overall, with lower shoulder muscle demand than HS in the lab dataset.



Implementation: Designate H22 as the preferred hammer for wall- and pole-height striking tasks and phase out HS for these task categories where practical.

- Use H22 Over HS for Floor-Level Tasks [HIGH PRIORITY]

Floor-level striking produced consistently high shoulder postural demand for all hammers in the laboratory. When floor striking is unavoidable, use H22 as the preferred option because it reduced shoulder/arm demand compared with HS while maintaining comparable task output.

Supporting evidence: All hammers exceeded the high-risk shoulder-flexion threshold at floor height, but H22 showed lower shoulder flexion and lower upper-arm muscle demand than HS in the lab condition.

Implementation: For floor-level striking that cannot be avoided, prioritize H22 and pair it with setup changes and task rotation to manage exposure.

- Convert Floor-Level Tasks to Wall-Height Tasks Where Feasible [HIGH PRIORITY]

The strongest risk-reduction opportunity identified in the lab is to reduce or eliminate floor-level striking by changing task setup/height. Where feasible, reconfigure the work so repetitive striking occurs closer to waist-to-chest height.

Supporting evidence: Floor-level striking produced the highest shoulder flexion across laboratory conditions, whereas wall-height setups reduced shoulder elevation substantially.

Implementation: Review floor-level striking steps during planning and use benches, staging, or other positioning aids to raise the strike point where practical.

Engineering Recommendations

- Handle Geometry Refinement for H16 [MODERATE PRIORITY]

In the laboratory, H16 showed consistently poorer ergonomic performance than the other hammers, including higher shoulder/forearm demands in some conditions. If H16 remains in consideration, focus design changes on handle geometry and balance to reduce upper-extremity loading.

Supporting evidence: H16 ranked lowest for effectiveness across laboratory conditions and showed higher shoulder/forearm muscle demand than H22 in multiple comparisons.

Implementation: If pursuing improvements to H16, prototype and test revised handle/balance options and re-check shoulder and forearm loading against baseline performance.

- Adjustable-Height Platforms for Floor-Level Fastening [HIGH PRIORITY]

Because floor-level striking drove the highest shoulder postural demands in the lab, engineering controls that raise the work target are the preferred solution. Use adjustable supports, staging, or positioning aids to bring repetitive striking into a more neutral shoulder posture.



Implementation: Make height-adjustable supports available for phases that otherwise require frequent floor-level striking, and train crews to set them to a comfortable working height.

Administrative Recommendations

- Structured Rotation Off Floor-Level Tasks [HIGH PRIORITY]

When floor-level striking cannot be eliminated, use task rotation to limit prolonged time in high shoulder-flexion postures. Rotation is a key interim control while engineering changes are planned or deployed.

Implementation: Rotate workers between floor-level striking and more neutral tasks to reduce cumulative shoulder loading.

- H22-Specific Striking Technique Training [MODERATE PRIORITY]

If crews transition to H22, provide basic technique coaching so workers rely on hammer momentum rather than excessive grip/arm effort. This helps preserve the ergonomic benefits observed in the laboratory.

Implementation: Provide short, hands-on coaching on relaxed grip and controlled swings when introducing H22, and reinforce during routine safety/ergonomics check-ins.

PPE Recommendations

- Elbow Support and Anti-Vibration Gloves for Extended Floor Tasks [MODERATE PRIORITY]

For workers with extended floor-level striking exposure, consider optional supportive PPE (e.g., light elbow support and/or anti-vibration gloves) as an added comfort measure where it does not interfere with grip or control.

Supporting evidence: Elbow postures during floor-level striking fell in a moderate range across all hammers, suggesting PPE may be helpful for comfort in higher-exposure scenarios.

Implementation: If PPE is introduced, confirm it does not reduce grip security, dexterity, or strike accuracy during typical work.

8.5 Hammer Design Selection Matrix

This matrix summarises evidence-based hammer selection guidance derived from biomechanical outcomes (EMG, ROM, impact force, vibration) measured across field (scaffold knob / wood surfaces) and laboratory (POLE, FLOOR, WALL) conditions with four scaffold hammers. Surface type and task setup are the primary drivers of upper-extremity loading; treat hammer selection as a secondary control. Cells are colour-coded from green (recommended) through amber (use with caution) to red (avoid).



Table 6 SELECTION GUIDE: Surface Type & Task Condition

Condition / Criterion	H16 Estwing 16 oz (Lighter)	H20 Estwing 20 oz (Medium)	H22 Estwing 22 oz (Heavier)	HS Stiletto 15 oz (Control)
> BY SURFACE TYPE				
Scaffold Knob / Fitting (metal-to-metal, hard, low-compliance)	Acceptable	Good	Best	Avoid
Wood (compliant — formwork / framing / nailing)	Best	Good	Good	Avoid
> BY TASK CONDITION				
Pole Adjustment / De-adjustment (POLE) Rotational & percussive scaffold connections	Good	Good	Good	Acceptable
Floor-Level Strike (FLOOR) Raise work height where feasible — high shoulder demand	Good	Good	Good	Acceptable
Wall / Elevation Strike (WALL) Mid-to-upper working position	Good	Good	Best	Acceptable

Table 7 PERFORMANCE METRICS COMPARISON

Condition / Criterion	H16 Estwing 16 oz (Lighter)	H20 Estwing 20 oz (Medium)	H22 Estwing 22 oz (Heavier)	HS Stiletto 15 oz (Control)
Muscle Efficiency (lower mean EMG = less muscular demand)	Good	Good	Best	Poor
Forearm Extensor Loading (lower = less fatigue & MSD risk)	Low	Low	Low	HIGH (32.8% MVC)
Peak Impact Force Delivery (higher = more effective single strike)	Moderate	High	High	Moderate
Strike Effectiveness (joint ROM per unit muscle effort)	Good	Good	Best	Moderate
Vibration Exposure (no meaningful differences found)	Similar	Similar	Similar	Similar
OVERALL RECOMMENDATION (primary use case summary)	Secondary (compliant surfaces)	Secondary (hard surfaces)	PRIMARY (all conditions)	Not Recommended (highest forearm load)

LEGEND [BEST] Best / Primary [GOOD] Good - Suitable [ACC] Acceptable [AVOID] Not Recommended [--] No Difference



Key findings & Notes

- Surface type is the primary driver of upper-extremity loading: striking a scaffold knob/fitting consistently requires higher muscle effort and larger joint ROM than striking wood (up to ~75% reduction in shoulder ABD/ADD ROM on wood).
- Floor-level striking (FLOOR) consistently increases shoulder flexion demands regardless of hammer type: raise the work/strike point where feasible to reduce sustained shoulder flexion.
- H22 (Estwing 22 oz) ranked highest for muscle efficiency (lowest mean EMG) and strike effectiveness (most joint motion per unit muscle effort) on the knob surface, and highest for effectiveness on wood.
- HS (Stiletto 15 oz) showed the highest forearm extensor activation on wood (mean 32.8% MVC; statistically significant vs. all other hammers, $p = 0.026$). Not recommended for high-volume tasks.
- H20 and H22 produced significantly higher peak impact forces on knob surfaces vs. HS and H16 ($p < 0.0001$), without meaningfully higher vibration transmission.
- In the laboratory, hammer differences were generally small: posture and task setup (working height, reach, target location) are the dominant determinants of shoulder/elbow loading.



9 Limitations

Best practices for research were followed in this laboratory and field-based, real-world evaluation of scaffold assembling tasks. Given the nature of the work, the following limitations should be taken into consideration when interpreting the results:

- Sample size and data coverage: Ten participants contributed lab-based data and ten participants contributed to field-based data. Statistical power may be insufficient to detect moderate hammer-effect sizes
- Laboratory environment: Controlled conditions may not fully capture field-level variability including substrate compliance variation, tool wear, worker fatigue across a full shift, and the influence of personal protective equipment.
- Short exposure duration: Strike trials captured brief exposure windows (typically 7–15 strikes per trial). Cumulative exposure effects over multi-hour work periods were not assessed.
- Surface simulation: Knob and wood surfaces in the laboratory setting approximated but may not perfectly replicate all mechanical characteristics of field surfaces (e.g., varying fastener depth, substrate thickness, height above ground).



10 Future directions

Future studies could build on the present findings in several important ways. First, expanding analyses using the DOSE database could explore potential differences associated with hammer type during scaffold work throughout the work shift, where posture, reach constraints, and stability may differ from laboratory or simulated conditions. The natural task variability observed in real field scenarios (e.g., overhead vs. waist-level striking, constrained foot placement, or dynamic balance requirements) would improve ecological validity. Additional factors such as task duration, cumulative exposure, and fatigue effects may further refine our understanding of tool–user interactions.

Importantly, future research should more explicitly examine the discrepancy between self-reported comfort and objective physical demands observed in this study. While HS was consistently reported as the preferred hammer in both laboratory and field settings, H22 demonstrated the most favorable physical demand profile, characterized by lower overall EMG activation across measured upper-extremity muscles. Note: HS showed marginally more favourable impact-force attenuation in the field dataset; future work should examine whether this difference is meaningful under sustained exposure. Given that relatively few statistically significant differences were observed between hammers overall, these findings suggest that perceived comfort may play a critical role in tool preference that is not fully captured by biomechanical metrics alone.



11 Conclusion

The convergent findings of the Field Study and the controlled Laboratory Study provide a robust, multi-method evidence base for ergonomic action. The key conclusions are:

- ★ H22 is the preferred hammer for knob/scaffold-surface tasks and for wall- and pole-height conditions, where it consistently showed the most favourable profile of low overall EMG demands and high task effectiveness. These advantages were consistent across field and laboratory settings; however, laboratory hammer differences were generally small relative to the effect of task setup and working height, and should be interpreted accordingly.
- Task height is a stronger determinant of shoulder flexion risk than hammer design. Floor-level striking universally exceeds the HIGH risk threshold ($>60^\circ$ ShFLEX) regardless of hammer used. Engineering controls are required.
- HS should be replaced as the standard-issue hammer for hard-surface and overhead tasks. It performs worst on forearm activation (Field, Wood), shoulder flexion (Lab, Floor), and task effectiveness.
- H16 should not be used on hard (Knob/scaffold) surfaces until its handle vibration amplification is addressed. It also produces the highest posterior deltoid demands across both studies.
- Statistical limitations (small sample sizes in both studies) are acknowledged, but the consistency of directional findings, risk threshold crossings, and effectiveness rankings across independent studies provides sufficient evidence for action.
- Comfort and preference should be considered alongside biomechanical metrics when selecting hand tools. Although H22 showed the lowest muscle demands (EMG activation), participants consistently rated HS as most comfortable in both laboratory and field settings; given the limited significant differences between hammers, subjective experience may meaningfully influence real-world tool preference.



12References

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